Funding Instruments and Effort Choices in Higher Education *

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Abstract

This paper examines the effects of Tuition-free Higher education (TFHE) policies on student enrollment and academic performance, with a focus on the 2016 Chilean reform that granted tuition-free higher education to students from the lowest five income deciles. Using a difference-in-differences approach, we find that TFHE increased enrollment and persistence in higher education on the eligible but had modest effects on graduation and dropout rates. To disentangle the role of student effort from selection effects, we develop a structural model in which students choose effort levels in response to financial incentives. Our results highlight that TFHE expanded access, in particular for low-achieving students. Despite the removal of academic progress requirements, we found no evidence of weakening performance.

Keywords: Financial aid, Effort choices, Higher Education
JEL Codes: I22, I23, I26

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1 Introduction

Higher education is a central determinant of long-run earnings, inequality, and intergenerational mobility. Despite decades of expansion in attainment, the returns to a college degree have remained large (Card, 1999; Goldin & Katz, 2008), and access to selective institutions is a powerful lever for upward mobility (Chetty et al., 2020). Yet completion is unequally distributed, and the gap in degree attainment between high and low-income youth has widened in recent decades (Bailey and Dynarski, 2011, Dynarski et al., 2023). In response, governments have invested heavily in policies aimed at lowering financial barriers to college. As of today, more than 40 countries operate some form of tuition-free higher education (TFHE) policy, generally limited to public institutions, and recent debates in the United States over student-debt forgiveness, free community college, and need-based grants reflect a similar emphasis. Public spending on higher education in the OECD has risen accordingly (OECD, 2019), raising the question of how aid should be designed to balance equity, fiscal cost, and educational outcomes.

Higher education policies operate on two margins. On the extensive margin, lowering net tuition draws in marginal students who would otherwise have stayed out. On the intensive margin, the design of aid impacts field choice, persistence and completion among those who do enroll. Traditional instruments combine financial relief with performance requirements to keep incentives aligned. The Pell Grant in the United States is conditioned on satisfactory academic progress, and merit-based scholarships typically require a minimum GPA to renew (Scott-Clayton, 2011, Deming and Dynarski, 2010). Subsidized loans expand access but burden graduates with debt that distorts post-college choices (Rothstein and Rouse, 2011, Sieg and Wang, 2018, Black et al., 2023). TFHE sits at one extreme of this design space, eliminating out-of-pocket tuition for a broad eligible population without imposing progress conditions, and thus removing both a financial barrier and a performance lever simultaneously. The ex-ante effect on educational outcomes is therefore ambiguous. Eliminating tuition draws into the system a less-prepared pool of marginal entrants while, for inframarginal students, it relaxes a binding budget constraint, frees up time previously devoted to part-time work, and reduces dropout after negative income shocks. Removing the performance requirement also impacts selection (drawing in students with a higher cost of effort) while reducing incentives for effort among inframarginals (Scott-Clayton, 2011). Countries with TFHE tend to exhibit lower aggregate graduation rates than comparable countries in which students face out-of-pocket fees, but aggregate measures cannot distinguish compositional changes from behavioral responses during college (Ferreyra et al., 2022).

This paper studies the effects of a large-scale TFHE reform on enrollment, persistence, and

academic performance, and develops a framework that disentangles selection from changes in effort. We exploit Chile's 2016 *Gratuidad* reform, which granted tuition-free higher education to students from the lowest five income deciles in roughly half of universities and vocational centers (including some private institutions), as a natural experiment, and proceed in three steps. First, we implement a difference-in-differences design to estimate effects on enrollment, persistence, dropout, and graduation. Second, we build and estimate a dynamic structural model of enrollment, program choice, and academic effort that endogenizes how students respond to changes in out-of-pocket tuition. We use the estimated model to simulate counterfactual designs that re-introduce performance-based conditions on top of the tuition-free benefit, and assess their implications for fiscal cost and educational attainment. The structural exercise allows to decompose aggregate impacts into a composition effect from marginal students induced to enroll and a behavioral effect on the effort choices of infra-marginal students.

The Chilean context offers an ideal setting for this analysis, thanks to both rich administrative data and a well-defined policy intervention. We assemble administrative records that link the universe of national-exam takers to higher education enrollment, scholarship assignment, course-by-course credit completion, dropout, and graduation outcomes. The *Gratuidad* reform is a sharp, large-scale policy change with a clearly defined eligibility rule based on income decile, introduced on top of a well-developed pre-existing aid system that included subsidized loans (the State Guaranteed Loan), need-based scholarships, and a centralized admission platform. Crucially, the pre-existing instruments were all performance-conditional, both at entry (scores at the National Entrance Exam) and during studies (course completion and GPA). *Gratuidad* replaced these conditional schemes with full, unconditional tuition coverage for eligible students. The reform therefore varied both the level of net tuition and the performance conditioning of aid, providing the variation needed to identify selection and effort responses separately.

The difference-in-differences estimates reveal a sizable extensive-margin response. Enrollment among eligible students rose by 3.5 percentage points (p.p.) in 2016 and stabilized at around 10 p.p. from 2017 onward, when the policy was extended to vocational centers. The increase is concentrated among lower-ability students who, prior to the reform, did not qualify for either subsidized loans or merit scholarships. Their enrollment rose by 14 p.p., compared with 5.5 p.p. for students already eligible for the most generous pre-existing aid. We find no evidence of crowding-out at non-eligible institutions, in contrast to the spillover patterns documented by Londoño-Vélez et al. (2020) in Colombia, suggesting that the bulk of the response comes from genuinely marginal students rather than substitution across in-

stitutions. As a result, the composition of incoming cohorts shifted to the left of the ability distribution, a pattern with first-order implications for downstream outcomes.

Estimating effects on educational outcomes is less straightforward, as the change in composition confounds the response of infra-marginal students with that of new entrants. We exploit a feature of the policy design to address this. Cohorts that had already enrolled before 2016 became eligible for *Gratuidad* on their remaining tuition payments, with treatment intensity varying by cohort but with enrollment decisions already made. This holds composition fixed and isolates the response of infra-marginal students. Among these students, the policy reduced dropout by up to 2.5 p.p. and increased on-time graduation by 1.8 p.p. These results suggest a net-positive effect of the two-component policy. For 2016-and-after cohorts, by contrast, we find essentially null effects on dropout, graduation, and persistence, pointing to a negative composition shift that is roughly offset by the policy bundle.

While the reduced-form analysis shows that the policy was successful in expanding access without lowering aggregate completion rates, three reasons motivate a structural approach. First, the difference-in-differences estimates cannot decompose, for each aggregate outcome, the relative contribution of marginal students induced to enroll and infra-marginal students who change their behavior. Second, the policy bundled two changes that the data cannot separately identify without further structure, namely the reduction in out-of-pocket tuition and the elimination of performance requirements. Third, the policy is fiscally costly. Between 2011 and 2019, public spending on higher education in Chile rose by 160% and reached 5.4% of GDP, against 2.9% on average in the rest of the OECD (OECD, 2019). Evaluating budget-improving alternatives that preserve the equity gains of *Gratuidad* requires a model in which students respond to changes in both prices and incentives.

For these reasons, we develop a dynamic discrete-choice model of enrollment, field and institution choice, dropout, and academic effort. In the first stage, a student observes her type (gender, ability, socioeconomic status and unobserved trait) and her personalized choice set (Fack et al., 2019), and decides whether and where to enroll. Conditional on enrollment, in each subsequent period she chooses whether to continue and how much effort to exert, where effort shapes the distribution of credit completion. Performance feeds back into next-period out-of-pocket tuition through the rules of pre-existing scholarship programs, generating a dynamic link between current effort and future financial incentives. Effort is unobserved and recovered from a first-order condition following De Groote (2025), which allows the marginal cost of effort to be estimated jointly with the fixed costs of enrollment. Identification of effort responses relies on the quasi-experimental variation in out-of-pocket tuition induced by the differential timing of the reform across cohorts, together with the performance-conditional

nature of the pre-existing aid system.

We estimate the model using the 2016–2017 entry cohorts, for whom we observe academic progression for up to seven years. Our sequential estimation procedure combines the conditional choice probability approach of Hotz and Miller (1993) with the recovery of effort costs from optimality conditions following De Groote (2025). We account for persistent unobserved heterogeneity using the two-step grouped fixed-effects approach of Bonhomme et al. (2022), classifying students into two latent types based on residual credit-completion patterns. Conditional on these types and observed characteristics, we estimate dropout probabilities and credit-completion distributions, recovering effective effort from the latter. These estimates, together with predicted lifetime earnings, allow us to construct continuation values without repeatedly solving the full dynamic model. We then substitute the effort first-order condition into the value of enrollment, estimate fixed costs by maximum likelihood, and recover marginal effort costs. Because estimation uses post-reform cohorts, the difference-in-differences estimates provide an external benchmark for assessing the model's simulated responses to the introduction of *Gratuidad*.

Estimation reveals substantial heterogeneity in costs across the student distribution. Fixed costs increase with out-of-pocket tuition, especially at the enrollment stage. They are also higher for female students, for students enrolling in a different region, and for students who have fallen behind in their program. Marginal costs of effort exhibit a clear gradient in both ability and income, with lower-PSU and lower-income students facing higher marginal costs of producing a unit of academic progress. Critically, we find no evidence that the removal of performance requirements weakened effort among infra-marginal students. The composition of new entrants induced by *Gratuidad*, who exhibit higher fixed and marginal costs, accounts for most of the post-reform progression patterns, while the behavior of incumbent students changes little. The model therefore rationalizes the reduced-form evidence. Aggregate completion is held in check by composition rather than by moral hazard.

We use the estimated model to simulate alternative funding designs aimed at reconciling the equity gains of *Gratuidad* with stronger incentives and lower fiscal cost. Three scenarios are central. Re-introducing an annual performance requirement (completing 70% of registered credits) to retain free tuition restores the conditional component removed by the reform and tightens within-cohort incentives to exert effort. Imposing a minimum PSU score of 500 for eligibility, which harmonizes TFHE with the threshold of pre-existing merit scholarships, reduces the number of beneficiaries while attracting better-prepared students. Combining both elements compounds the selection and incentive effects. The performance requirement raises dropout among students with high marginal costs but improves progression among

those who remain. The PSU requirement is fiscally lighter and improves graduation rates through selection. The joint design preserves a large share of *Gratuidad*'s equity gains while substantially reducing the cost per graduate. We further discuss design variants that target early-career liquidity, differentiate thresholds across academic and vocational programs, or replace hard cutoffs with a smooth eligibility index.

Related literature This paper builds upon several strands of literature. It first contributes to the literature on higher education financing and student outcomes. Broad reviews emphasize that aid design affects not only access, but also persistence, completion, and the timing at which students learn about prices and eligibility (Deming and Dynarski, 2010, Page and Scott-Clayton, 2016, Dynarski, 2003). There is substantial evidence showing that increasing grants or reducing net tuition raises enrollment and attainment (Dynarski, 2003, Castleman and Long, 2016, Angrist et al., 2015, Denning, 2017, Fack and Grenet, 2015, Londoño-Vélez et al., 2020, Dobbin et al., 2022, Harris and Mills, 2025). Comparative evidence from Solis, 2017, Molina and Rivadeneyra, 2021, Duryea et al., 2023, and Bolhaar et al., 2024 further shows that the effect of subsidies depends on the pre-reform financing environment and on whether aid takes the form of grants or loans. Evidence on completion and performance is more mixed once composition effects are taken into account (Cohodes and Goodman, 2014, Denning, 2019, Denning et al., 2019, Ferreyra et al., 2024). Beyond partial equilibrium responses, Epple et al., 2006, Epple et al., 2017, and Abbott et al., 2019 show that equilibrium supply responses and redistribution matter for the incidence of education-finance reforms. This paper complements that literature by studying a large-scale tuition-free reform in a setting where the public aid system was already well developed, and by tracing both extensive-margin enrollment responses and post-enrollment outcomes.

It also contributes to the growing literature on tuition-free college and the design of free-college programs. Bucarey (2017) studies the crowding-out effects of Chile's tuition-free reform on low-income students at selective programs. Recent work emphasizes that free-college design is consequential. Unconditional or early-committed aid can generate larger take-up responses than aid that remains uncertain or application-intensive (Burland et al., 2023, Harris and Mills, 2025), while dynamic models imply that free college often raises enrollment more than graduation because outcomes depend on the type of students induced to enroll and on their effort choices (Matsuda, 2020, Ferreyra et al., 2024). Comparative evidence from promise programs and tuition reversals also documents meaningful heterogeneity across institutional settings (Carruthers and Fox, 2016, Schoonover et al., 2026, Bietenbeck et al., 2023). Our contribution is to bring this design question to the Chilean reform and to quantify how much of the observed effect operates through selection versus changes in effort

among infra-marginal students.

Second, we contribute to the literature on the incentive structures and moral hazard inherent in educational funding. Traditional subsidies lower the cost of entry, but they may also weaken the marginal cost of delay or failure. Scott-Clayton (2011) shows that academic incentives embedded in aid rules affect college achievement. Garibaldi et al. (2012) and Beneito et al. (2018) show that continuation tuition and retake pricing can significantly reduce delay and increase effort, while Montalbán (2022) finds that cash support is most effective when accompanied by demanding progress requirements. Related evidence on aid packaging and borrowing frictions (Marx and Turner, 2018, Falco and Reichlin, 2025) suggests that students respond both to the level and to the composition of aid. TFHE also releases students from debt accumulation, which has been shown to affect major choice (Rothstein and Rouse, 2011), dropout (Stinebrickner and Stinebrickner, 2008), career decisions (Sieg and Wang, 2018), and post-college household formation (Black et al., 2023). This paper speaks directly to that mechanism because the Chilean reform simultaneously changed out-of-pocket tuition and removed academic-progress requirements for a large group of students.

Finally, we contribute to a growing structural literature on educational choices under dynamic incentives. Larroucau and Rios (2024) study retention and reapplication decisions in a centralized admissions market in which students learn about their match quality over time. Dynamic models have also been used to study the design of financial aid. Colas et al. (2021) derive and quantify optimal need-based aid in a life-cycle model of enrollment, dropout, and parental transfers, while Joensen and Mattana (2024) compare grants and loans in a model of enrollment, work, and borrowing in which merit requirements condition aid on credit accumulation. In both, academic progression is a stochastic process that does not respond to policy. A smaller set of papers model effort explicitly. Ferreyra et al. (2022) treat in-college effort as an input to class completion in Colombia and show that omitting it overstates the role of ability, while Tincani et al. (2026) and Arslan (2026) study pre-college effort responses, to preferential admissions in Chile and to tuition subsidies in Turkey, the former allowing for biased beliefs about the returns to effort. Finally, De Groote (2025) studies track choice and grade retention in secondary school, where he shows that accounting for effort responses is key to predict changes in graduation and college enrollment. We borrow De Groote (2025) identifcation of marginal costs of effort changes in student financing in higher education, where effort determines eligibility for aid and the cost of continuing. We contribute by linking current effort to future educational costs through out-of-pocket fees and performance requirements, which allows the model to decompose adverse selection from moral hazard under a large-scale tuition-free reform.

Overview The rest of the paper is organized as follows. section 2 describes the institutional details, the TFHE policy and introduces the data. section 3 reports results from differences-in-differences estimation, and section 4 builds a model of enrollment and effort in higher education. section 5 discusses the main results of the model and section 6 performs counterfactual simulations. section 7 provides a discussion and next steps.

2 Background

This section describes the institutional setting, the policy of interest, and the data used in our analysis. subsection 2.1 provides an overview of the Chilean higher education system and the pre-existing funding instruments. subsection 2.2 describes the Tuition-Free Higher Education policy introduced in 2016 and the substitution patterns it generated across the previously available aid programs. subsection 2.3 introduces the administrative records that we use to track enrollment, performance, and graduation outcomes, and defines the samples used in the reduced-form and structural analyses.

2.1 Institutional Setting

Chilean higher education is well developed, with enrollment rising steadily since the General Law of Universities (*Ley General de Universidades*) of 1981, which incentivized the creation of institutions by allowing entry without state dependence. In 2023, the system comprised a total of 138 institutes: 80 vocational schools (33 Professional Institutes and 47 Centers of Technical Formation) and 58 universities.[1]

Universities are divided between the so-called *traditional* universities (a mix of public and private institutions that receive direct support from the government) and *private* universities, which constitute the rest. The latter were created after 1981 and are mainly financed through tuition. Traditional universities are officially known as the Universities of the Rector's Council (*Consejo de Rectores de las Universidades Chilenas, CRUCH*) and are responsible for coordinating the higher education system. They comprise 30 universities and account for around half of total enrollment.[2]

Access to higher education is through a Centralized Admission Platform, compulsory for traditional universities and some private institutions. Vocational program admission is also possible, although most are conducted off-platform. After high-school completion, students

[1] https://www.ayudamineduc.cl/ficha/instituciones-vigentes-reconocidas-por-el-mineduc.

[2] In 2018, the government established a clear set of rules for being considered a traditional university with the promulgation of Law 21091, allowing the first three non-traditional universities to enter.

applying to higher education take the centralized admission exam (*Prueba de Selección Universitaria*, PSU). This standardized test, similar to the SAT in the United States, ranges from 150 to 850 points, with a mean of 500 and a standard deviation of 110. Students then submit rank-ordered lists of up to 10 degree programs through the centralized system and are assigned via a deferred-acceptance algorithm (Gale & Shapley, 1962). In practice, the platform imposes a minimum PSU score of 450 to apply.

Higher education is costly in Chile. In 2009, the average tuition fee for a university program was equivalent to 47% of the median family income (Solis, 2017). Costs vary across institution types but remain high even in public institutions. Different funding instruments coexist. Students rely mainly on loans and grants from the Ministry of Education. Eligibility criteria are strongly related to PSU scores and family income, as summarized in Figure A4.The State Guaranteed Loan program (SGL), introduced in 2006, finances 90% of reference tuition and requires a PSU score of 475 or higher, with no socioeconomic requirement since 2014. Several scholarships also exist, with the *Beca Bicentenario* and *Beca Excelencia* being the most popular. On average, scholarships finance 80% of reference tuition. Eligibility requires a PSU score of 500, and excludes students in the top three income deciles. Both short-cycle programs (SCPs) and university degrees are covered by scholarships. Two-year programs typically require a high-school GPA above 5.0. Access to private loans is limited: in 2015, only 7.5% of student loans came from banks without a state guarantee (Bucarey et al., 2020).

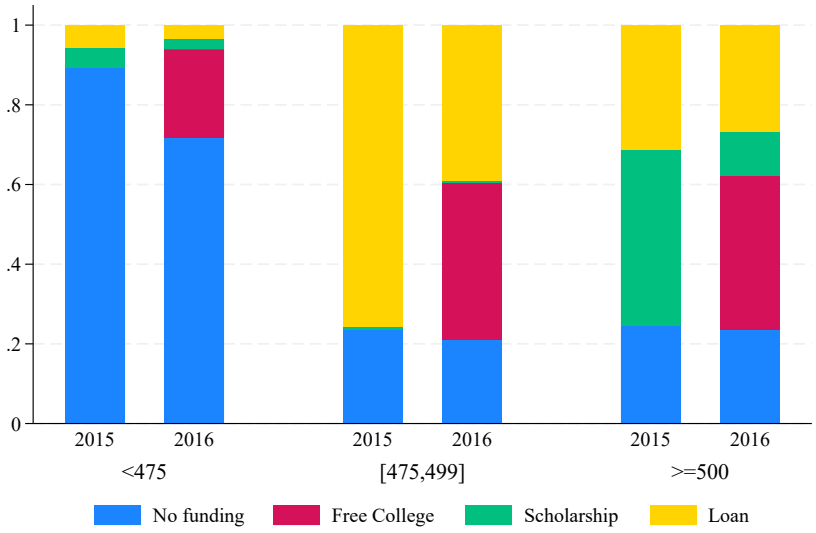
2.2 Tuition-Free Higher Education (TFHE) program

Since the implementation of SGL, student debt rose at a rate of 70% annually, and the number of recipients increased steadily from 15,800 in the first year to 652,000 students in 2016 (Bucarey et al., 2020). In 2011, students led mass protests demanding more affordable higher education. Michelle Bachelet was elected president in 2014 on a platform that included making college free by 2020. In 2015, the TFHE law was enacted, removing tuition fees for students in the bottom half of the income distribution. The policy was first implemented for the 2016 university cohort, expanded in 2017 to include vocational institutes, and further extended in 2018 to the sixth income decile. Applications for TFHE, loans, and scholarships are made through the Centralized Admission Platform by completing a short online form during the application period, regardless of whether the applicant applies to a platform degree or not. A schematic visualization of the policy is provided in Figure A5.

The introduction of TFHE policy generated different substitution patterns depending on students' test scores. Figure 1 shows the distribution of funding instruments before and

after the policy, by test score segment. Students scoring below 475 were ineligible for subsidized loans or merit-based scholarships (except for a small scholarship targeting top students from low-SES schools). For this group, TFHE reduced the share of students with no funding. Students scoring between 475 and 500 qualified for subsidized loans but remained below the threshold for most scholarships; for them, substitution was mainly from loans to TFHE, relieving them of any debt upon graduation. Students scoring above 500 qualified for merit-based scholarships (conditional on income eligibility). For this group, substitution was mainly from scholarships to TFHE, which had two advantages: first, scholarships typically covered only 80% of tuition and fees, whereas TFHE provided full funding; second, scholarships required students to meet satisfactory progression standards (validating about 70% of annual course credits), while TFHE imposed no such requirements. Before the policy, about 20% of scholarship beneficiaries lost eligibility; of these, around 40% switched to subsidized loans, while the rest continued without public funding. After the policy, most substituted to TFHE, while those above the income threshold continued either with loans or no funding, in roughly the same proportions as before.

Figure 1: Distribution of funding instruments, by test score



Notes: This figure shows the distribution of funding instruments by test score group, for 2015 and 2016. PSU is the national standardized test. Students with test scores below 475 are not eligible for public funding, students scoring between 475 and 500 are eligible for loans, and students scoring above 500 are eligible forscholarships.

2.3 Data

We use Chilean administrative records that provide detailed student and program data, including off-platform private universities and SCPs. We observe enrollment, admission test

scores, scholarship assignment, socioeconomic information, and demographic characteristics. Institutional data include type, tuition, location, and program length.In addition, we employ novel data on credit completion from the Information Service of Higher Education (SIES) covering the universe of enrolled students from 2016 to 2023. These data report the number of courses registered and whether they were completed.

Quasi-Experimental sample We restrict the sample to first-time test takers for the 2012 to 2017 cohort. Since we observe the data until 2023 and the standard university program lasts five years, we allow seven years for students to complete their program. Table B1 reports enrollment, controls, entry funding and outcomes for this sample by cohort. Between 100,000 to 120,000 students take the national exam for the first time each year, of whom about 60% enroll in higher education, split roughly half and half between the universities and SCPs. Panel B show that around 55% of students are female. Panel C show student financial aid. In 2013, around 30% of student received a subsidized loan at some point during their studies, and 40% received scholarship. In 2016, these shares diminished to 19% and 25% respectively, while 34% of students received TFHE, which increased up to 59% in 2020. In 2012, scholarships were expanded to include the third income quintile, such that all eligible students for TFHE were already eligible for scholarships conditional on scoring a certain PSU score (Bucarey, 2017).

Model sample For the structural model, we also exploit the rich data on credit completion. This dataset is only available from 2016. Together with allowing students 7 years for graduation, we estimate the model on a random sample of 2016-2017 cohorts. Descriptive statistics for the model sample are reported in Table B2. It is a random sample of 7,000 students per cohort, which is representative of the full population (panels A, B and C). Panel D shows that students register almost all required credits per year (93%) which is consistent with the fact that tuition is paid annually irrespective of the registered courses. Students complete around 80% of the annual curriculum, which represents about 85% of the courses they register.

3 Policy Effects

This section provides reduced-form evidence on the effects of the TFHE policy on enrollment and educational outcomes. subsection 3.1 presents the differences-in-differences specification and discusses the identifying assumption. In subsection 3.2, we estimate the extensive-margin response and document substantial heterogeneity across ability groups and institu-

tion types. subsection 3.3 addresses the composition problem that arises when estimating intensive-margin effects, and isolates the response of infra-marginal students by exploiting the differential exposure of pre-2016 cohorts to the policy. subsection 3.4 discusses the main threats to identification and presents robustness checks.

3.1 Empirical Strategy

We are interested in causally estimating whether the TFHE policy had an impact on enrollment and educational outcomes. To do so, we exploit the exogenous variation in eligibility introduced by the policy and estimate it s effect by differences-in-differences, in its Two-Way Fixed Effects framework. We estimate the following regression model by OLS:

$$Y_{it} = \sum_{\substack{k=2011 \\ k \neq 2015}}^{2020} \beta_k \left(\mathbb{1}\{t = k\} \times \mathbb{1}\{dec_i < 6\} \right) + \gamma \, \mathbb{1}\{dec_i < 6\} + \delta_{\mathbf{t}} + \mathbf{X}_{\mathbf{i}}' \alpha_{\mathbf{x}} + \epsilon_{\mathbf{it}} \tag{1}$$

Y_{it} is an outcome of interest including enrollment and educational outcomes such as dropout, graduation, on-time graduation and persistence, defined as the time (in years) spent in college. $\mathbf{X}_{\mathbf{i}}'$ includes a constant, gender, PSU score, high school GPA, degree type, and mother education. $\mathbb{1}\{dec_i < 6\}$ determines eligibility to TFHE, defined as belonging to the lower half of the income distribution. δ_t are cohort fixed effects. We report heteroskedasticity-robust standard errors. We use the year before the implementation (2015) as the reference category. β_t is the differences-in-differences estimator.

The identifying assumption is that, in the absence of treatment, eligible and non-eligible students would have experienced similar trends in the outcomes of interest ($\beta_k < 2015$). We test for the validity of this assumption by examining pre-treatment trends. The design also requires no violation of the stable unit treatment value assumption (SUTVA): the outcomes of a student must not depend on the treatment status of other students, so there must be no spillovers between eligible and non-eligible students (for instance, through changes in the composition of applicants or in the availability of seats at institutions). subsection A.1 reports descriptive evidence on these margins: admission cutoffs and tuition by institution tier (Figure A8 and Figure A9),[3] and vacancies (Figure A10). Admission cutoffs show no

[3]Elite: Pontificia Universidad Católica de Chile, Universidad de Chile, Universidad de Concepción, Universidad Técnica Federico Santa María, Pontificia Universidad Católica de Valparaíso and Universidad de Santiago de Chile. G8: Universidad Adolfo Ibáñez, Alberto Hurtado, Andrés Bello, de los Andes, del Desarrollo, Diego Portales, Finis Terrae and Mayor. Other CRUCH: the remaining universities of the Council of Rectors. Off-plat: universities outside CRUCH that are not in the G8. SCP: short-cycle programs at any institution.

visible break around the introduction of TFHE, tuition keeps evolving in line with its pre-existing trend, and vacancies did not expand much at adhering institutions, consistent with the 2016 Budget Law, which capped the growth of first-year enrollment at participating institutions at 2.7% unless authorized by the Ministry of Education (DIPRES, 2016).

Other papers have also estimated the causal effect of TFHE. Since family income is only observed for students who apply for aid through the FUAS, Martin (2024) uses the FUAS income decile and models aid applications jointly with college choice, as the policy increased applications. He also estimates a LATE with a fuzzy regression discontinuity at the eligibility cutoff, and Clasing-Manquian (2022) estimates a difference-in-differences with predicted eligibility. We complement these approaches by classifying students with the family income reported at the time of the PSU, before any aid application.

SUTVA cannot be tested directly, so the evidence in subsection A.1 is suggestive rather than conclusive: it may not capture displacement across programs or changes within institutions. As an additional safeguard, we restrict the sample to the bottom and top income deciles, dropping students close to the eligibility threshold, who are the most exposed to displacement. This restriction also sidesteps the "disappearance of the seventh decile" (Gezer, 2026), the distortion in enrollment among students just above the threshold.

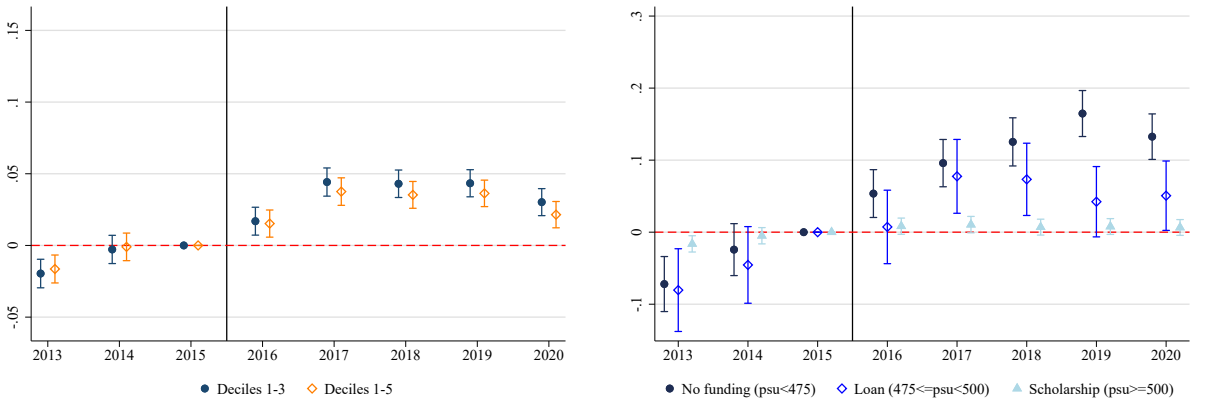
The differences-in-differences estimator can be interpreted as the intent-to-treat effect (ITT) of the policy, given that not all eligible students actually receive TFHE (some do not enroll, while others enroll in non-eligible institutions).

3.2 Effects on enrollment

We first examine the impact of TFHE on the extensive margin, i.e., enrollment. The 2016 cohort is the first for which the policy could affect the enrollment decision. Figure 2 shows the results of estimating Equation 1. As shown in panel a), enrollment among students in the bottom three income deciles increased by 1.7 p.p. in 2016 and by around 4.3 p.p. from 2017 to 2019, relative to 2015, compared to students in the top two deciles. The effect for the bottom five deciles is slightly smaller (1.5 p.p. in 2016 and 3.5 to 3.8 p.p. from 2017 to 2019). This increase in enrollment comes from students for whom the returns to education became positive or who were previously credit constrained. The larger effect, starting in 2017, is consistent with the expansion of free higher education to SCPs. As explained in section 2, the control group are students in the upper income deciles. We then examine heterogeneity by ability groups, which determine loan and scholarhip eligibility. In panel b), we observe that lower-ability students, who had limited access to public funding before 2016,

are the group that increases enrollment the most. Differences between groups are significant in 2017, when lower-ability students increase enrollment by 9.6 p.p., compared to 1.0 p.p. for students who already had access to merit scholarships.

Figure 2: Enrollment

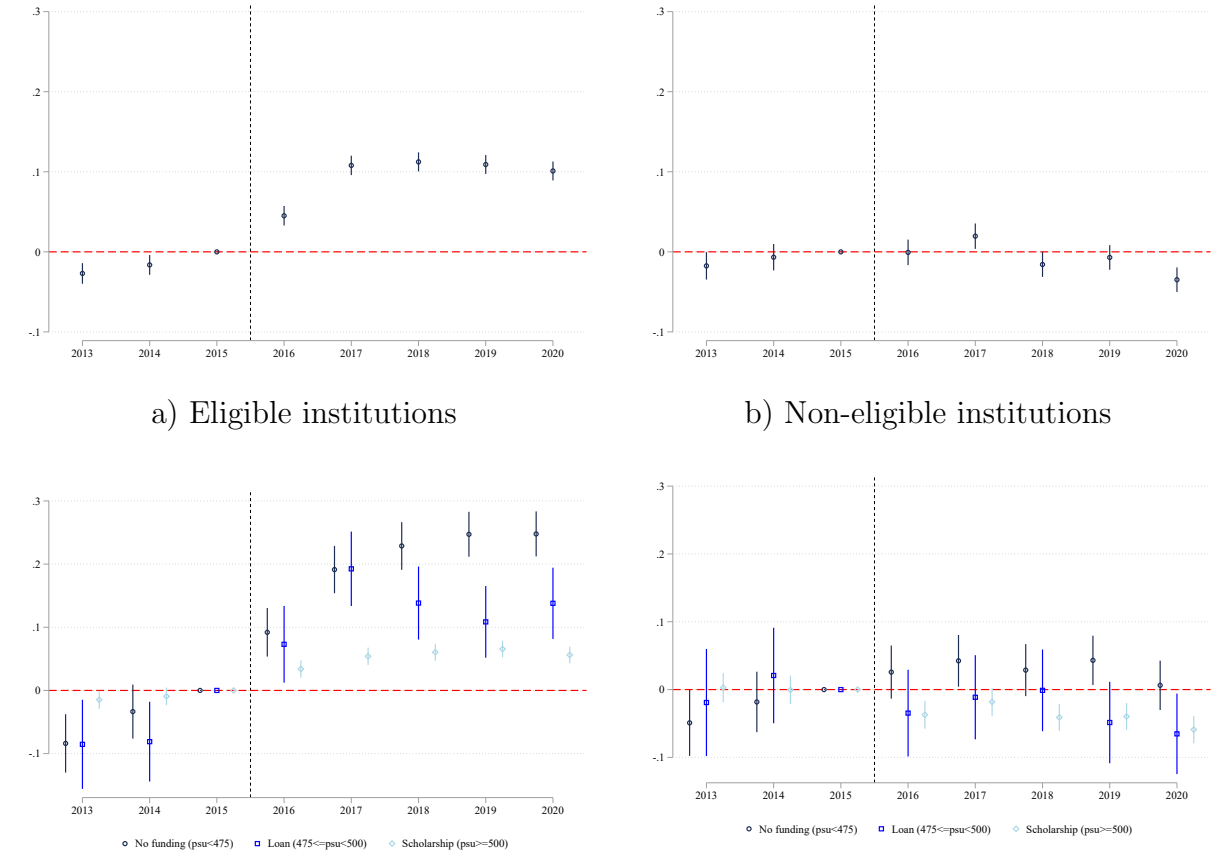


a) Effect on enrollment b) Effect on enrollment by funding eligibility

Notes: These figures show the results of estimating Equation 1. Sample is the universe of first time test takers. Panel (a) shows the aggregate results, comparing students in income deciles 1-3 (and 1-5) with students in deciles 9-10, using family income self-reported at PSU registration, while panel (b) estimates the equation for students in deciles 1-3 by groups based on their test scores. PSU is the national standardized test. Students with test scores below 475 are not eligible for public funding, students scoring between 475 and 500 are eligible for loans, and students scoring above 500 are eligible forscholarships.

Substitution between eligible and non-eligible institutions Since TFHE affected around 50% of institutions and 70% of enrollment, it is also important to examine whether there was substitution from non-eligible to eligible centers, or whether enrollment is driven completely by marginal students. Figure 3 reports the estimation on enrollment for eligible and non-eligible institutions. Results show a significant increase of eligible students to higher educations institutes that were eligible to the policy. In panel a), we observe increases of 4.5 and 10.8 p.p. for years 2016 and 2017, respectively. In contrast to Londoño-Vélez et al. (2020), panel b) suggests no crowding-out. In fact, we find mostly a null effect on enrollment (except for 2017). When looking at panel c), we observe that the increase in enrollment in eligible institutions is driven by low-ability students. Panel d) decomposes the null effect on noneligible institutions in a decrease in enrollment coming from high-ability students while low-ability students increase enrollment around 2 to 4 p.p. A possible explanation is that the policy generated awareness about higher education opportunities and low-ability students that had enough resources opted for higher education even if not eligible to TFHE.

Figure 3: Enrollment by institution eligibility



a) Eligible institutions

b) Non-eligible institutions

c) Eligible institutions by funding eligibility

d) Non-eligible institutions by funding eligibility

Notes: These figures show the results of estimating Equation 1 on enrollment for eligible and non-eligible institutions to the TFHE policy. Sample is the universe of first time test takers. Panel a) and b) show the aggregate results, while panels c) and d) disaggregate results by test score group.

Field Preferences We next look at whether students' preferences changed as a result of the policy. We use students' rank-ordered list and construct the dependent variable as the share of a field in student's list. We restrict the sample to eligible institutions and estimate Equation 1 for each field. Results are presented in Figure A7. Panel a) shows that TFHE increased preferences for Health by 1.5p.p. in 2016 and 2p.p. in 2017, while decreasing preferences Arts and Humanities by around 2p.p. STEM majors also lose popularity starting in 2018 with respect to 2015 by 2 p.p. Panel b) decomposes the results into finer categories. The decreases in Arts and Humanities are driven by Education, which decreases by 1p.p., 10% with respect to 2015 average. These results match the evidence found by Castro-Zarzur et al. (2022), which finds that TFHE eclipsed an incentive program to attract talent in

Education majors [4]. The increase in Health preferences can be explained by the fact that they are the most expensive programs, along with law majors. Finally, the the 2p.p. decrease in STEM majors is driven by Technology.

3.3 Effects on educational outcomes

Recovering treatment effects for the intensive margin is not as straightforward. If we were to estimate educational outcomes using Equation 1, composition and moral hazard could difficult interpretation: the distribution of students' test scores shifted to the left starting 2016, given that a substantial fraction of lower-ability enrolled (marginal students) as can be seen in Figure A2. On top of that, 2016 cohort onwards eligible students were not subject to performance requirements anymore. To keep the composition of students fixed, we exploit a particularity of the policy: cohorts prior to 2016 were eligible to TFHE for the remaining years of the nominal length of their degrees. For example, if they enrolled in 2015 in a five-year degree, they can be exempted of tuition for the remaining four years, two years for the 2015 cohorts, etc.

We can use this exogenous variation to partial out selection. 2012 to 2015 cohorts are potentially eligible to TFHE starting 2016 but cannot change their initial enrollment decision by construction. We adapt Equation 1 as

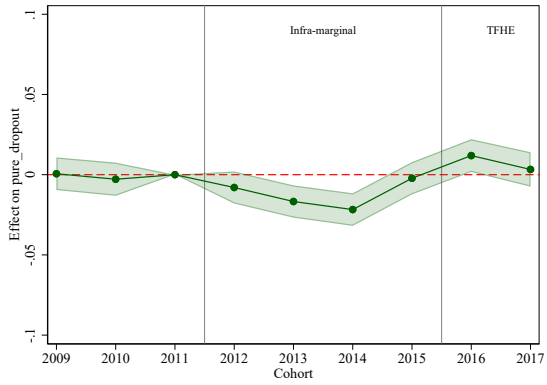
$$Y_{it} = \sum_{\substack{k=2010 \\ k\neq 2011}}^{2017} \beta_k^c \left(\mathbb{1}\{t=k\} \times \mathbb{1}\{dec_i < 6\}\right) + \gamma^c \, \mathbb{1}\{dec_i < 6\} + \delta_t^c + \mathbf{X_i}'\boldsymbol{\alpha}_x^c + \epsilon_{it}^c \tag{2}$$

The above regression remains a standard TWFE differences-in-differences regression, as the time dimension are cohorts. Treatment varies in intensity by cohort, but effects are always estimated with respect to a never treated cohort, so there are no issues of negative weighting (Callaway et al., 2024; de Chaisemartin & D'Haultfœuille, 2024). The variation exploited is purely cross-sectional.

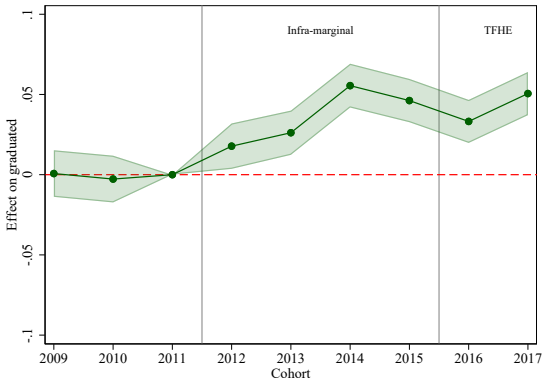
Figure 4 shows the results of estimating Equation 2. Panels a) to d) show the results for dropout, graduation, years enrolled, and graduation on time, respectively, for students in deciles 1-3 relative to deciles 9-10 (the estimates for deciles 1-5 are very similar). We find that dropout decreases by up to 2.2 p.p. for infra-marginal students (2014 cohort), and the effect reverses for the 2016 cohort (+1.2 p.p.), which is no longer different from zero in 2017. The composition of new entrants seems to dominate the positive effect from infra-marginal students. Graduation rises by up to 5.6 p.p. for students that receive the subsidy after

[4] see Pal (2026) for a comprehensive analysis of this policy.

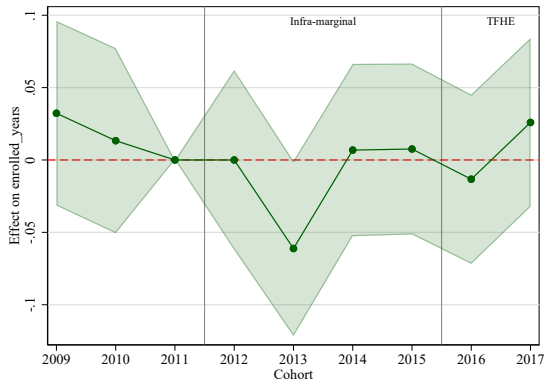
Figure 4: DID outcomes



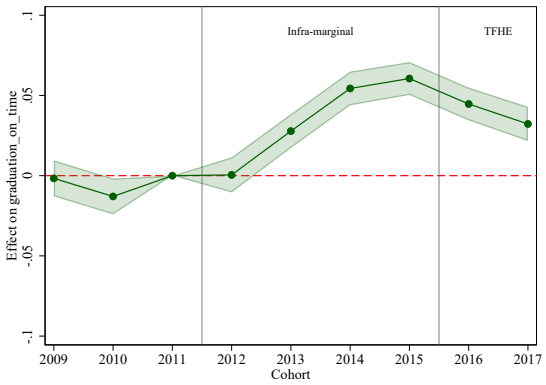
a) Dropout



b) Graduation



c) Years enrolled



d) Graduation on Time

Notes: These figures show the results of estimating Equation 1 on dropout, graduation, years enrolled, and graduation on time. Treated students are those in income deciles 1-3 (and 1-5), and control students are those in deciles 9-10, using family income self-reported at PSU registration. Sample is enrolled students. Cohorts 2012 to 2015 are potentially eligible to TFHE starting 2016 but already made their initial enrollment decision. Cohort 2011 is the reference category.

their enrollment decision (2014 cohort), but remains 3.3 and 5.1 p.p. above the 2011 cohort for the 2016 and 2017 cohorts, so we cannot rule out that this pattern reflects trends in the pre-policy cohorts. Years enrolled is imprecisely estimated and shows no clear pattern. Finally, as the subsidy gets larger (2014 and 2015 cohorts), the proportion of students who graduate on time rises by 5.4 and 6.1 p.p., and the effect declines to 4.5 and 3.2 p.p. for the 2016 and 2017 cohorts.

These results are defined as the overall effect of the policy on those who already enrolled. TFHE lowers the cost of staying enrolled, which tends to reduce dropout and increase graduation. It does so by lowering out-of-pocket and the removal of performance requirements to students who previously benefited from scholarships. Conversely, students with lower PSU scores that now enter higher education face more difficulties to progress and graduate. Quasi-experimental findings suggest that both forces are at play and cancel each other out. The total effect is unsignificantly different from zero, meaning the total number of graduates is larger (same graduation rate, but more students). To validate reduced-form results and quantify both effects, we next build a model of enrollment and major choice with endogenous effort.

3.4 Discussion and robustness

Our quasi-experimental analysis suggests that TFHE policy had a positive effect on enrollment on eligible students, especially for low-ability students that previously had limited access to public funding. We find no evidence of crowding-out of non-eligible institutions, suggesting that most of the increase in enrollment comes from marginal students. Eligible institutions were able to absorb the increase in enrollment, especially lower-quality institutions. Regarding educational outcomes, we find modest effects on dropout and graduation for infra-marginal students, while there are no significant effects for students that enrolled after the implementation of the policy, despite being of lower observed ability on average.

There are some limitations to our empirical strategy. We used eligibility to TFHE as a proxy for treatment, and compared eligible to non-eligible students as we do not have a continuous measure of income.

Crowd-out of average ability, high-income students If high-income students were to reduce their enrollment as a result of the policy, our estimates would be biased upwards. However, spillovers are unlikely as cut-offs scores were mostly unchanged, suggesting that the policy did not put additional stress on the capacity of institutions. As a robustness check, we estimate Equation 1 restricting the sample to income deciles 1, 2 and 9, 10, as crowd-out

is more likely to occur around the eligibility threshold, and find similar results.

Increase in test-takers If the policy induced more low-income students to take the PSU test, our control group would be positively selected, biasing our results upwards. However, we find no evidence of an increase in test-takers as a result of the policy. This is because take-up rates were already high (around 85% of students in their last year of high school take the test). In fact, what we see is an increase in higher education applications from low income students.

Income manipulation Income is self-reported and verified by the goverment. Misreporting could lead to students that are actually ineligible being classified as eligible. If this is done on purpose, it could bias our estimates upwards.

4 Model

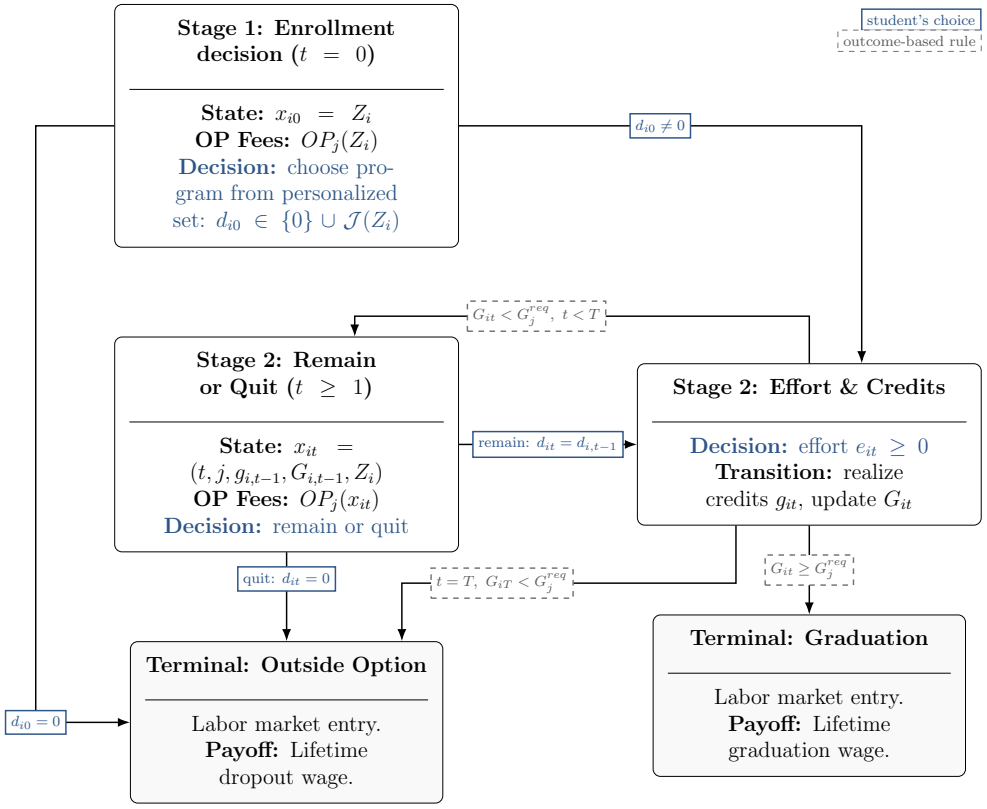
The decision process of an individual considering attending higher education can be depicted in two stages. In the first stage, she chooses whether to enroll, and if so, which field–institution combination to attend. To decide, she observes program characteristics and available funding instruments, and optimizes how much effort to exert in each program in her personalized choice set, determined by her score in the national exam (Fack et al., 2019).[5] Effort choice will determine her distribution of performance outcomes (credits). Conditional on enrollment, performance today endogenously determines out-of-pocket fees for the subsequent year. The student graduates when she cumulates the required number of credits of the enrolled program.

4.1 Timing and choices

Figure 5 describes the timing and choices of the model. A student is characterized by her type Z_i, defined by sex, ability (psu and gpa), socioeconomic status and unobserved trait m. In stage 1, the student decides whether to enroll in higher education. She observes her type Z_i, which characterizes her personalized choice set $\mathcal{J}(Z_i)$ and the funding instruments available. She decides between a field–institution combination $d_{i0} = j$, with $j \in \mathcal{J}(Z_i) = \{1, \ldots, J(Z_i)\}$

[5] Choice sets are student specific: a field–institution combination is feasible for a student in a given year only if her PSU score is at least the program's admission cutoff for that year. For programs admitting through the centralized platform, the cutoff is the score of the last applicant admitted in the first round; for off-platform programs, it is the lowest PSU score among first-year enrollees. Short-cycle programs impose no score requirement and are therefore available to all students. Higher-scoring students thus face strictly larger choice sets, and the outside option is always available.

Figure 5: Structural model layout



a program from her personalized choice set, and the outside option $j = 0$, where she would directly earn her expected lifetime non-enrollment wage.

Stage 2 is conditional on enrollment. It is an optimal stopping problem.[6] The student decides in each subsequent period whether to continue in the program, $d_{it} = \{d_{i,t-1}, 0\}$, and how much effort $e_{it} \in [0, +\infty)$ to exert. By exerting effort, the student implicitly chooses the probabilities of performance outcomes $g_{i,t+1}$, which accumulates over time as $G_{it} := \sum_{s=1}^{t} g_{is}$.

The model allows for $T = 7$ academic periods.[7] They are indexed by $t = 1, \ldots, T$. The

[6]The model abstracts from switching programs after enrollment: a student who leaves her program is treated as a dropout. We do so for three reasons. First, data availability: separating switching from non-switching requires following students over more periods than our panel provides, so that a switch can be distinguished from a spell of non-enrollment and a student's post-switch outcomes can be compared with those of stayers. Second, our difference-in-differences estimates show that the policy effect operates mainly through the extensive margin and through performance and persistence within the initial program, so switching is not a first-order margin for the effects we study. Third, switching is more naturally rationalized by models in which students learn about their match or ability over time, as in Larroucau and Rios (2024), whereas our model focuses on effort and financial incentives. Treating switches as dropouts therefore affects the level of measured attrition, but not the mechanism of interest.

[7]Since a typical degree program lasts five years, this horizon builds in a two-year margin, allowing

nominal length of program j is $\bar{t}_j$. A student can choose to drop out before the terminal boundary and earn the expected lifetime dropout wage. Dropout wages vary with the year of exit, capturing returns to incomplete higher education. In the final academic period she still exerts effort and can graduate; after that performance realization, a nongraduate exits mechanically because an eighth enrollment period is outside the model.

During periods before graduation ($t = 1, \ldots, \bar{t}_j - 1$), performance affects both the probability of dropout and, for scholarship holders, out-of-pocket fees $OP_j(x_{it})$: failing to meet the performance requirement ($g_{it} < 3$) implies paying the full tuition the following year. During the graduation period ($t = \bar{t}_j, \ldots, T$), the student can graduate and obtain the corresponding lifetime expected wage, postpone graduation when $t < T$, or choose to drop out. In period T, effort and graduation remain endogenous, but a nongraduate is forced to exit after performance is observed. Hence, the state space is $x_{it} = (t, d_{i,t-1}, g_{i,t-1}, G_{i,t-1}, Z_i)$.

4.2 Utility and the dynamic program

Flow utility from being enrolled in program j during period t includes individual preferences for different programs, out-of-pocket fees sensitivity, and the psychological or physical "disutility" of academic work. Following De Groote (2025), we decompose it into a fixed cost (FC_j) that captures preferences for higher education, and a variable cost of effort:

$$u_j(x_{it}) + \varepsilon_{ijt} = -FC_j(x_{it}) - c_j(x_{it})e_{it} + \varepsilon_{ijt}, \tag{3}$$

with marginal cost $c_j(x_{it}) > 0$. Fixed costs $FC_j(x_{it})$ summarize all non-effort related preferences for programs including out-of-pocket fees $OP(Z_{i,}, g_{i,t-1}, P_{j,t})$, defined as the fraction of program fees not covered by the state for student i in program j. These depend on tuition $P_{j,t}$, type Z_i and past performance $g_{i,t-1}$:

$$OP(Z_{i,}, g_{i,t-1}, P_{j,t}) = \begin{cases} (1 - \lambda(Z_i, g_{i,t-1}))P_{j,t} & \text{if the student holds a scholarship,} \\ (1 - \lambda(Z_i))P_{j,t} & \text{otherwise.} \end{cases}$$

where $\lambda(\cdot) \in [0, 1]$ is the subsidy rate, i.e. the fraction of program fees $P_{j,t}$ covered government. For scholarship holders, $\lambda(\cdot)$ depends on past performance g_{it-1} as well as on

completion to be measured with delay. The average program duration is 6.5 years for an undergraduate, 5 years for Professional Insitutes and 3 years for technical programs (SIES, 2026).

individual characteristics Z_i, since failure to meet credit requirements results in the loss of the subsidy. For loans and TFHE, $\lambda(\cdot)$ depends only on observable characteristics and does not vary with performance. Finally, the distribution of the idiosyncratic shock $\varepsilon_{ijt} \sim$ EV1 is iid and common knowledge. It captures what the student learns at the beginning of the period.

The dynamics of the optimization problem are the following: On the one hand, by exerting more effort today, the student i) increases her probability of accumulating credits and eventually graduating in subsequent periods and therefore raises her expected future wage, and ii) increases her probability to maintain public funding (passing the performance threshold) if eligible. On the other hand, exerting effort is costly and reduces the flow utility of attending higher education. The conditional value function $v_j(x_{it}, e_{it})$ can be written as

$$v_j(x_{it}, e_{it}) + \varepsilon_{ijt} = u_j(x_{it}, e_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it})\, \bar{V}(x_{i,t+1}(\bar{g})) + \varepsilon_{ijt}, \tag{4}$$

where $\bar{V}(x_{i,t+1}(\bar{g}))$ denotes the ex-ante (continuation) value function evaluated at the post-performance state $x_{i,t+1}(\bar{g})$. The first term corresponds to the current flow utility from enrolling in field–institution j. The second term captures the expected continuation value, discounted by β. Uncertainty in $x_{i,t+1}(\bar{g})$ arises solely from the realization of the performance outcome $\bar{g}$.

Given the Extreme-Value Type 1 assumption on the utility shock, we have a closed-form solution for the ex-ante value function, commonly know as the logsum expression, where γ is the Euler-Mascheroni constant. Instead, when the choice (choosing the outside option) or the outcome leads to a deterministic state, the student earn either the present expected lifetime expected value of either graduating or dropping out. Summarizing, the continuation value is given by:

$$\bar{V}(x_{i,t+1}(\bar{g})) = \begin{cases} \gamma + \ln\Big(\sum\limits_{j' \in \mathcal{J}(Z_i)} \exp v_{j'}\big(x_{i,t+1}(\bar{g})\big) \Big) & \text{if } t \leq T+1 \text{ and } G_{it} < G_j^{req}, \\ \alpha_w\, wage_j(x_{i,t+1}) & \text{if } G_{it} \geq G_j^{req}, \\ \alpha_w\, wage_0(x_{i,t+1}) & \text{if } t = T+1 \text{ and } G_{it} < G_j^{req}, \\ & \text{or if } d_{it} = 0. \end{cases} \tag{5}$$

4.3 *Effective* effort and performance

Effective effort e_{it} is unobserved, while performance outcomes g_{it} are directly recovered from students' records. We model performance outcome $g_{i,t+1}$ as the result of effort e_{it} and a logistically distributed shock η_{it}:

$$g_{i,t+1} = \kappa \quad \text{if } \bar{g}_{jt}^{\kappa} < \ln(e_{it}) + \eta_{i,t+1} \leq \bar{g}_{jt}^{\kappa+1}, \tag{6}$$

where the cutoff values $\bar{g}_{jt}^{\kappa}$ and $\bar{g}_{jt}^{\kappa+1}$ are field–institution–year specific. In practice, performance is discretized into five categories, $\kappa \in \{0, 1, 2, 3, 4\}$. Students observe the realization of the shock $\eta_{i,t+1}$ only after choosing their effort e_{it}, but since they know the distribution of $\eta_{i,t+1}$ (standard logistic), they can compute the probability of each outcome conditional on effort:

$$\Pr\left(g_{i,t+1} = \kappa \mid e_{it}, x_{it}\right) = F\big(\ln(e_{it}) - \bar{g}_{jt}^{\kappa}\big) - F\big(\ln(e_{it}) - \bar{g}_{jt}^{\kappa+1}\big)\,, \tag{7}$$

where $F(\cdot)$ denotes the logistic CDF. The higher the effort, the higher the expected performance. By choosing effort, students choose the distribution of performance outcomes. Making use of the logistic assumption of the performance shock $\eta_{i,t+1}$, for $\kappa = 0$ we can also rewrite Equation 7 as:

$$e_{it} = \frac{1 - \Pr(g_{i,t+1} = 0 \mid e_{it}, x_{it})}{\Pr(g_{i,t+1} = 0 \mid e_{it}, x_{it})}. \tag{8}$$

Effective effort can thus be interpreted as the odds of avoiding the lowest performance outcome (failing all registered credits).

Students observe realized performance outcomes $g_{i,t+1}$, which determine their state and therefore whether the student is forced out, is allow to conitnue in the program or graduates. The model allows for heterogeneous performance thresholds $\bar{g}_{jt}^{\kappa}$ across programs j, capturing differences in grading standards and program requirements. Dropout or graduating entails receving the expected lifetime wage of a nongraduate or graduate, respectively.

Both lifetime expectedwages are computed as the discounted sum of expected annual wages

over a working life of 40 periods, a common discount factor β:

$$wage_\ell(x_{it}) = \sum_{s=0}^{39} \beta^s \, w\big(s,\ \text{female}_i,\ \text{region}_\ell,\ field(\ell),\ \text{inst_type}(\ell)\big), \qquad \ell \in \{j, 0\}, \tag{9}$$

where s denotes years of labor market experience and $w(\cdot)$ is the expected annual wage, a function of experience, gender, region of residence, field and institution type. The same function applies to graduates ($\ell = j$) and to dropouts ($\ell = 0$); the two differ only in the field and institution-type arguments, which for dropouts take their nongraduate values. Its empirical specification is given in Equation 11.

4.4 Solution of the model

The model is solved by backward induction. Higher education is no longer feasible in $T+1$ after the seventh performance realization, or once the student completes all required program credits, as detailed in Equation 5.

The terminal continuation value is used as an input in earlier periods. Importantly, performance in period T still changes graduation and therefore the relevant wage; only the subsequent enrollment/dropout choice is absent.

At each period t, the student first chooses an effort level e_{it} in her program j. The decision trades off a loss in flow utility with higher future expected gains. Assuming she is rational and behaving optimally, she is solving the following first-order condition for each program [8]:

$$\frac{\partial v_j(x_{it}, e_{it})}{\partial e_{it}} = \frac{\partial u_j(x_{it}, e_{it})}{\partial e_{it}} + \beta \sum_{g \in \mathcal{G}} \frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}} \bar{V}(x_{it+1}(\bar{g})) = 0 \quad \text{if } e_{it} = e_j^*(x_{it}). \tag{10}$$

At the current-period state x_{it} the ex-ante value function $\bar{V}$ can always be computed, either by the logsum formula when a choice is involved, or directly the lifetime expected wage, as described in Equation 5, and the model is solved recursively by iterating backward to $t = 1$. Given the set of optimal effort levels $\{e_j^*(x_{it})\}_{j \in \mathcal{J}(x_{it})}$, the student chooses the program with the highest value $v_j(x_{it}, e_j^*(x_{it}))$. The resulting choice probabilities take the familiar logit

[8] a sufficient condition for an interior solution is that students believe there is a non-zero probability to avoid the lowest performance outcome. Positive marginal costs ensure an infinite amount of effort is never exerted.

form:

$$p_{jt} = \frac{\exp\big(v_j(x_{it}, e_j^*(x_{it}))\big)}{\sum_{j' \in \mathcal{J}(x_{it})} \exp\big(v_{j'}(x_{it}, e_{j'}^*(x_{it}))\big)}.$$

4.5 Identification

As is standard in the dynamic discrete choice literature, the discount factor is not separately identified from the per-period utility function (Magnac & Thesmar, 2002). We therefore fix β at 0.95 and normalize the utility of the outside option to the lifetime expected dropout wage.

We further assume that not enrolling in higher education is an absorbing state: individuals who drop out cannot resit the entrance exam or re-enter higher education in a later period. This restriction rules out option-value effects from delayed re-entry and lets the outside option be summarized by a single terminal payoff.

4.5.1 Identification of unobserved types

The unobserved component of Z_i is a latent student trait $m \in \{1, 2\}$. We assume that students are drawn from a mixture of two types, with type probabilities $\pi_m(Z_i^{obs})$ that may depend on observed initial characteristics. The type is fixed over the student's life cycle, known to the student, and unobserved by the econometrician. It shifts the utility, effort and performance primitives, so that students with identical observed states may behave differently because of their type.

We rely on the identification arguments for dynamic discrete choice models with time-invariant unobserved heterogeneity of Arcidiacono and Miller (2011); see also Arcidiacono and Ellickson (2011). First, conditional on the observed state x_{it} and the type m, choices, performance outcomes, and state transitions are independent across periods, so that the trait is the only source of persistence in unobservables. Second, the transition of the observed state depends on the type only through the observed state and the current choice, and the trait itself does not change over time. We have enough periods of choices and outcomes for the mixture to be identified from the within-student serial dependence in choices: the type-specific conditional choice probabilities and the type probabilities π_m are then jointly identified from the joint distribution of choice sequences, up to a labeling of traits. Under these conditions, the type enters the CCP-based representation of the continuation value exactly as an observed state would, which allows us to estimate the model with the conditional-choice-probability approach used below, replacing the unobserved type by its

posterior probability given the observed history (Arcidiacono & Ellickson, 2011; Arcidiacono & Miller, 2011).

4.5.2 Fixed and Marginal costs

Marginal costs c_j are identified from the FOC (De Groote, 2025). The model attributes higher performance levels, conditional on the same continuation value, to lower marginal costs of effort. Fixed costs then rationalize the observed enrollment shares across programs. The sensitivity to out-of-pocket fees is pinned down by the variation in public funding eligibility, after controlling for observables, which serves as a price shifter.

Wage elasticity α_w is identified through an exclusion restriction. We use local labor market field-region variation. The underlying assumption is that conditional on region and field fixed effects, the interaction provides idiosyncratic local demand shocks that shift utility only through the wage. Region enters the model only through the wage, that is, it is excluded from utility. The region-by-field wage variation (estimated in subsection 4.6) therefore shifts utility only through the wage and identifies α_w.

4.6 Estimation

We are able to circuvent full solution methods by representing the future value term as the conditional value function and a non-negative term that depends on the empirical conditional choice probabilities (Hotz & Miller, 1993). Given the Type-1 extreme value assumption on utility shocks, CCPs are of logit type. We specify flexible parametric functional forms for CCPs, effective effort, fixed costs and wage regression. Details on the estimation of these components are provided in Section Appendix C.

Because effort solves the first-order condition Equation 10, all students sharing the same state x_{it} exert the same effort $e_j^*(x_{it})$, so the index $\ln(e_j^*(x_{it}))$ in Equation 6 is a function of the state alone. We therefore estimate performance directly as a flexible ordered logit of credits (discretized into five categories, 0–4) on the state, with thresholds that vary by field, institution and year.

Wages are estimated as

$$\begin{aligned}\log(w_j(x_{it})) = \beta_0^w + \beta_{f,field(j)}^w \,\text{female}_i + \beta_e^w \,\text{exper}_{it} + \beta_{e2}^w \,\text{exper}_{it}^2 \\ + \sum_{r=1}^{16} \beta_{r,field(j)}^w \, R_{ir} + \beta_{inst_type(j),\,field(j)}^w + \varepsilon_{ijt},\end{aligned} \tag{11}$$

where R_{ir} is a dummy for region r and inst_type$(j) \in \{\text{SCP}, \text{Private}, \text{CRUCH}\}$ denotes the institution type of program j. The gender gap and the institution-type premium are both allowed to vary by field, and the region-field interactions $\{R_{ir}\, \beta^{w}_{r,field(j)}\}$ provide the variation used to identify α_w: conditional on region and field fixed effects, they act as local demand shocks that shift utility only through the wage. Because the wage equation is estimated on survey data, which is not linked to students' higher education history, the unobserved type cannot be accounted for in wages.

Exiting higher education is a terminal action, so $v_0(x) = \alpha_w wage_0(x)$. Hotz and Miller (1993) show that the continuation value can be inverted as a function of the outside option continuation value and a log correction term, which corresponds to the conditional dropout probability

$$\bar{V}(x_{i,t+1}(\bar{g})) = \gamma + \alpha_w wage_0(x_{i,t+1}) - \ln p_{0,t+1}(x_{i,t+1}(\bar{g})). \tag{12}$$

Replacing Equation 12 in Equation 5, we can write the continuation value as a function of the data, where the dropout probability $p_{0,t+1}$ is estimated as a flexible logistic regression of the observed dropout choice on the state.

From the first-order condition in Equation 10, we can directly recover the expression for marginal costs:

$$c_j(x_{it}) = \beta \sum_{\bar{g} \in \mathcal{G}} \frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}} \bar{V}(x_{it+1}(\bar{g})) \quad \text{if } e_{it} = e^*_j(x_{it}), \tag{13}$$

where $\bar{V}(x_{it+1}(\bar{g}))$ is computed using the estimated CCPs and dropout wage, and $\frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}}$ can be directly computed given the logistic assumption about η_{t+1} in Equation 6.

Replacing the expression for marginal costs in the flow utility, and defining

$$a^{\bar{g}}_{ijt} = \phi^{\bar{g}}(e^*_{ijt}) - e^*_{ijt} \left. \frac{\partial \phi^{\bar{g}}(e_{ijt})}{\partial e_{ijt}} \right|_{e_{ijt}=e^*_{ijt}}.$$

Note that the continuation component can be split as

$$EMAX_{ijt} = \sum_{\bar{g}\in\mathcal{G}} a_{ijt}^{\bar{g}} \left[\gamma - \ln p_{0,t+1}(x_{i,t+1}(\bar{g}))\right],$$
$$EWAGE_{ijt} = \beta \sum_{\bar{g}\in\mathcal{G}} a_{ijt}^{\bar{g}} wage_{\bar{g}}(x_{i,t+1})$$

to estimate α_w. $wage_{\bar{g}}$ equals the graduation wage on graduation branches and the dropout wage otherwise. At the terminal period T, $EMAX_{ijt} = 0$ by construction, while $EWAGE_{ijt}$ remains nonzero and preserves the effort incentive operating through the probability of graduation.

Finally, we reached the following expression for the conditional value function that we can directly estimate from the data

$$v_j(x_{it}, e_{ijt}^*) = -FC_j(x_{it}) + \beta EMAX_{ijt} + \alpha_w EWAGE_{ijt}. \tag{14}$$

Estimation algorithm: We estimate the model jointly with an expectation-maximization (EM) algorithm in the spirit of Arcidiacono and Miller (2011), in which the posterior probability q_{im} that student i has unobserved trait m is updated using both performance and choices. The trait distribution $\pi_m(Z_i^{obs})$ is a population prior conditional on observed characteristics and is defined for every student in the structural sample, including those who never enroll, who inform it through their enrollment decision. The algorithm proceeds as follows; the full description is in subsection C.4 of the appendix.

1. **Wages.** We estimate the wage equation Equation 11 on survey data and predict graduation and dropout wages for every program-state. Since the survey is not linked to students' higher education histories, wages do not depend on the unobserved trait and are estimated once, outside the EM loop.

2. **Starting values.** We start from posteriors q_{im} that increase with students' average performance, and check that the converged estimates do not depend on this choice by repeating the estimation from alternative starts.

3. **EM loop.** Iterate over the following steps until the posteriors, the parameters and the objective stop changing:

   (a) *Reduced-form components.* Weighting each student by q_{im}, we estimate the ordered logit of credit completion and the dropout logit, both with trait shifters. The dropout probabilities deliver the continuation values $\bar{V}(x_{i,t+1}(\bar{g}))$ through

Equation 12, and hence $EMAX_{ijt}^m$, $EWAGE_{ijt}^m$ and $c_j^m(x_{it})$ through Equation 13, separately for each trait.

(b) *Fixed costs, utility parameters and trait prior.* Given the continuation values, we maximize the mixture likelihood of performance and choices over the fixed-cost and utility parameters, including FC_j and α_w, and over the parameters of $\pi_m(Z_i^{obs})$.

(c) *Posterior updating.* We update q_{im} as the probability of trait m given the student's observed characteristics, performance and choices.

4. **Marginal costs.** Once the algorithm has converged, we compute $c_j(x_{it})$ from the FOC Equation 13 using the final CCPs, performance estimates and posteriors.

5 Results

This section first presents reduced-form estimates that predict dropout probabilities and performance outcomes. Then, structural parameters are discussed, which include students fixed and marginal costs.

5.1 Dropout probabilities

Table 1 reports the results of a logistic regression model of dropout. Overall, dropout probabilities depend strongly on the enrollment length. Progressing through the program reduces the likelihood of dropping out (relative to period 1). Being female reduces the probability of dropping out, with the effect becoming stronger as students progress in their studies.

Observed ability, measured as GPA during high school and exam entry score (PSU), is strongly correlated with persistence in higher education, but conditional on the student's progress (accumulated credits), its estimated coefficient diminishes. Both cumulative credits and the number of credits completed in the previous year are strong determinants of student progression. Being delayed in the program increases the probability of dropping out, with a stronger magnitude for females.

Type 2 capture students that are very sensitive to performance, and have a higher probaility of dropping out when being delayed and out-of-pocket fees are high.

Table 1: Dropout probabilities

	dropout	
OP fees	0.101***	(0.006)
Rel. Ability	-0.105***	(0.023)
Diff. Region	0.106***	(0.026)
Credits left	-0.113***	(0.004)
Designed total credits	0.099***	(0.006)
Delay	1.098***	(0.051)
Performance: category 0	2.116***	(0.040)
Performance: category 1	1.435***	(0.040)
Performance: category 2	0.291***	(0.033)
Performance: category 4	-0.721***	(0.049)
female	0.197**	(0.079)
Middle SES	0.033	(0.098)
High SES	0.007	(0.100)
PSU	0.063	(0.066)
GPA	-0.111**	(0.053)
Type 2	-10.276***	(0.656)
OP × female	-0.010**	(0.005)
OP × middle SES	-0.030***	(0.007)
OP × high SES	-0.027***	(0.008)
OP × psu	-0.026***	(0.004)
OP × gpa	0.015***	(0.003)
Delay × female	0.057	(0.050)
Delay × middle SES	0.040	(0.062)
Delay × high SES	0.233***	(0.062)
Delay × PSU	-0.191***	(0.035)
Delay × GPA	-0.062*	(0.034)
Delay × SCP	0.584***	(0.042)
t=2 × SCP	0.446***	(0.030)
t=3 × SCP	0.832***	(0.047)
t=4 × SCP	0.988***	(0.087)
t=5 × SCP	0.830***	(0.184)
t=6 × SCP	0.910**	(0.391)
Performance 0 × female	-0.301***	(0.039)
Performance 0 × middle SES	-0.060	(0.046)
Performance 0 × high SES	-0.093*	(0.051)
Performance 0 × PSU	-0.022	(0.027)
Performance 0 × GPA	-0.011	(0.026)
Performance 1 × female	-0.278***	(0.041)
Performance 1 × middle SES	-0.024	(0.048)
Performance 1 × high SES	0.053	(0.053)
Performance 1 × PSU	0.005	(0.027)
Performance 1 × GPA	0.072***	(0.027)
Performance 2 × female	-0.099***	(0.035)
Performance 2 × middle SES	-0.028	(0.043)
Performance 2 × high SES	0.082*	(0.046)
Performance 2 × PSU	-0.103***	(0.024)
Performance 2 × GPA	0.079***	(0.024)
Performance 4 × female	-0.328***	(0.052)
Performance 4 × middle SES	-0.046	(0.065)
Performance 4 × high SES	0.263***	(0.065)
Performance 4 × PSU	-0.180***	(0.036)
Performance 4 × GPA	-0.091**	(0.035)
Credits left × female	0.001	(0.002)
Credits left × middle SES	0.001	(0.003)
Credits left × high SES	-0.007**	(0.003)
Credits left × psu	0.007***	(0.002)
Credits left × gpa	0.010***	(0.002)
Rel. Abil. × female	0.038***	(0.011)
Rel. Abil. × middle SES	0.011	(0.013)
Rel. Abil. × high SES	0.057***	(0.015)
Rel. Abil. × psu	0.011*	(0.006)

5.2 Performance

Table B4 presents the results of the ordered logit model of performance with five categories (0–4). Coefficients follow a similar pattern as in the logistic dropout model. Ability is a strong predictor of performance, as is income. Out-of-pocket fees have a negative effect on performance, which is stronger for low-income and low-ability students.

Being delayed in the program reduces the probability of obtaining credits, and the effect is stronger for females. Finally, past credits is a strong predictor of current performance, particularly for low-ability students.

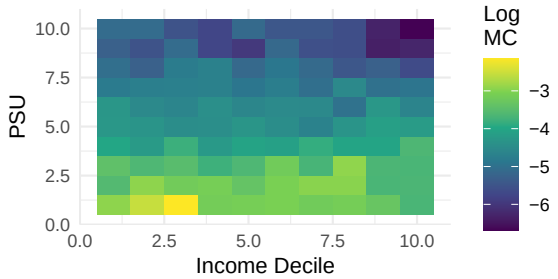
Type 2 students are worse performers, but they are hurt less than type 1 from being delayed and from out-of-pocket fees.

5.3 Fixed Costs

Estimates of fixed costs and expected wage are presented in Table 2. Students' fixed costs increase with out-of-pocket fees, particularly at the program choice stage. Once enrolled, this effect is substantially attenuated in the continuation decision. High SES students are less sensitive to out-of-pocket fees than their low SES counterparts, while the opposite is true for high-ability students. Students also face higher fixed costs when the program is located in a different region.

Past progression significantly affects fixed costs, as reflected in the coefficient of the delay variable. Again, females face higher fixed costs than males when delayed, and students who performed well on the national exam appear to experience smaller penalties from lagging behind. Expected wage has a positive effect on the conditional value function, indicating that students value their future expected earnings.

Figure 6: Marginal Costs



Notes: Heatmap of log marginal costs by PSU (y-axis) and income (x-axis) decile.

Figure 8 shows the distribution of log marginal costs by income and PSU score. The

Table 2: Fixed costs

Fixed Costs		
Enrol HE	2.959***	(0.144)
Enrol HE × Elite	-0.815***	(0.098)
Enrol HE × Other CRUCH	0.276***	(0.081)
Enrol HE × G8	0.370***	(0.122)
OP fees	0.024***	(0.001)
OP × low SES	0.025*	(0.013)
OP × middle SES	-0.002	(0.013)
Diff. Region	0.935***	(0.061)
Diff. Region × female	-0.178***	(0.051)
Diff. Region × low SES	0.417***	(0.068)
Diff. Region × middle SES	0.414***	(0.072)
Diff. Region × psu	0.197***	(0.034)
Diff. Region × gpa	0.221***	(0.034)
Relative ability	0.494***	(0.062)
Rel. Abil. × female	0.066***	(0.022)
Rel. Abil. × low SES	-0.426***	(0.028)
Rel. Abil. × middle SES	-0.265***	(0.030)
Performance (t-1): category 0	2.670***	(0.130)
Performance (t-1): category 1-2	-0.064	(0.092)
Performance (t-1): category 4	-0.968***	(0.087)
Credits (t-1) × female	-0.179***	(0.024)
Credits (t-1) × low SES	-0.024	(0.032)
Credits (t-1) × middle SES	0.005	(0.034)
Credits (t-1) × psu	-0.042**	(0.017)
Credits (t-1) × gpa	0.018	(0.016)
Credits left (t-1)	0.090***	(0.011)
Credits left (t-1) × female	0.018**	(0.008)
Credits left (t-1) × low SES	-0.048***	(0.011)
Credits left (t-1) × middle SES	-0.039***	(0.011)
Credits left (t-1) × psu	-0.052***	(0.006)
Credits left (t-1) × gpa	-0.005	(0.005)
Type 2	-1.067***	(0.044)
EMax	0.950	(.)
Expected wage (USD million)	23.826***	(1.139)
Expected wage × low SES	2.820**	(1.268)
Expected wage × middle SES	2.935**	(1.383)
Product FE	Yes	

Standard errors in parentheses

Fixed Costs and expected wage

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

heatmap exhibits a gradient in marginal costs in both ability and socioeconomic status dimensions.

Table 3 presents, for interpretational purposes, the unit-weighted projection of log marginal costs over the complete reachable state space. The fitted marginal cost decreases with PSU score and high-school GPA and increases with out-of-pocket fees and prior delay. The interaction coefficients allow these gradients to vary with student characteristics.

Table 3: Marginal costs

	log(mc)	
female	-0.408***	(0.062)
Low SES	-0.372***	(0.079)
Middle SES	-0.605***	(0.084)
psu	-0.306***	(0.060)
gpa	-0.214***	(0.046)
Type 2	3.314***	(0.016)
OP fees	-0.019***	(0.006)
Diff. Region	0.220***	(0.024)
Credits completed (t-1)	-0.781***	(0.013)
Credits left (t-1)	0.032***	(0.005)
t = 2	3.129***	(0.039)
t = 3	3.459***	(0.038)
t = 4	2.998***	(0.041)
t = 5	2.426***	(0.050)
t = 6	3.423***	(0.056)
t = 7	4.067***	(0.154)
Credits (t-1) × female	-0.017**	(0.007)
Credits (t-1) × low SES	0.023***	(0.009)
Credits (t-1) × middle SES	0.048***	(0.010)
Credits (t-1) × psu	0.063***	(0.006)
Credits (t-1) × gpa	-0.026***	(0.005)
Credits left (t-1) × female	0.000	(0.003)
Credits left (t-1) × low SES	0.017***	(0.004)
Credits left (t-1) × middle SES	0.017***	(0.004)
Credits left (t-1) × psu	-0.008***	(0.003)
Credits left (t-1) × gpa	-0.010***	(0.002)
OP × female	0.015***	(0.004)
OP × low SES	0.040***	(0.007)
OP × middle SES	0.025***	(0.006)
OP × psu	-0.009***	(0.003)
OP × gpa	-0.011***	(0.003)
Diff. Region × female	-0.008	(0.019)
Diff. Region × low SES	-0.125***	(0.025)
Diff. Region × middle SES	-0.097***	(0.027)
Diff. Region × psu	-0.016	(0.014)
Diff. Region × gpa	-0.020	(0.014)
Observations	19,606	
Field FE	Yes	
Institution FE	Yes	

Standard errors in parentheses

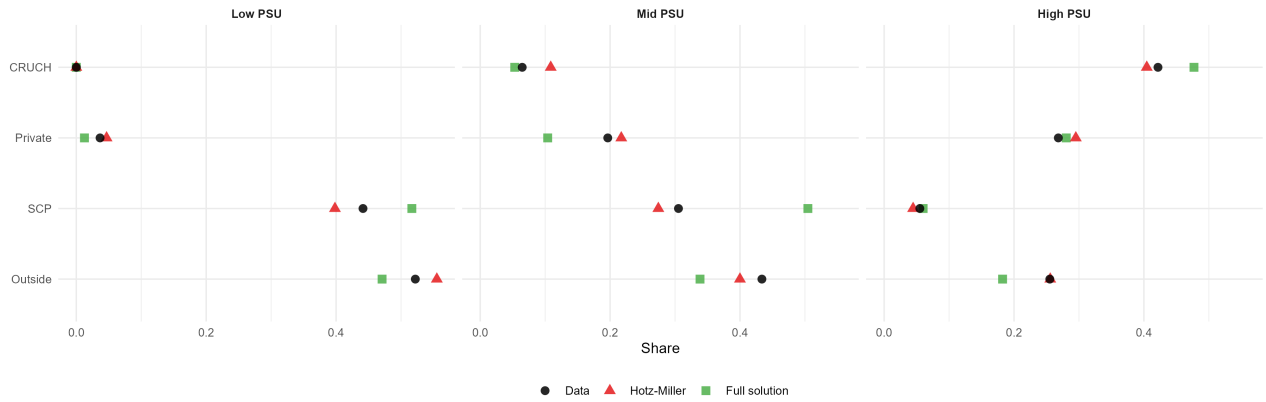
MC heterogeneity.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.4 Model Fit

5.4.1 In-sample Validation

Figure 7: Fit by ability tercile



Notes: In sample fit by PSU tercile. Dynamic model with endogenous effort.

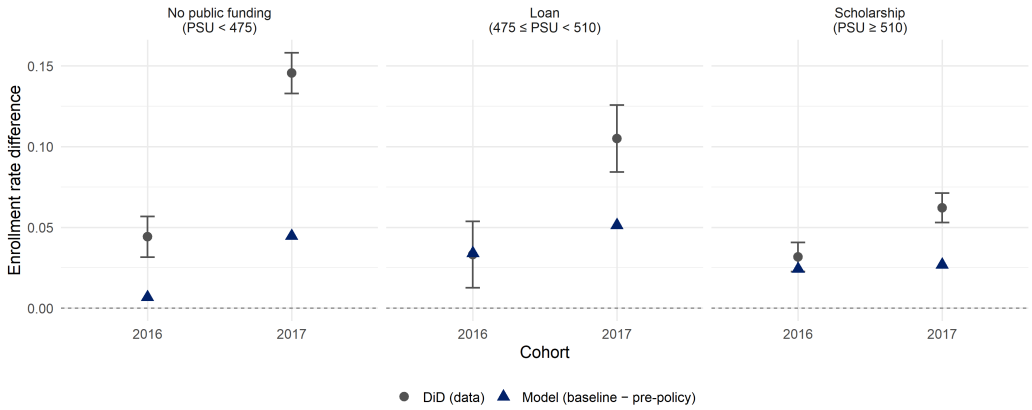
5.4.2 Out-of-sample Validation

The structural model is estimated entirely on post-reform (2016–2017, *Gratuidad*-era) data, so its parameters are never directly disciplined by pre-reform behavior. To check that the model is not simply curve-fitting the post-reform scenario, we use it to simulate a *pre-policy* counterfactual that reverts the funding rule to the regime that prevailed before 2016: the *Gratuidad* coverage rate is set to zero, so eligible students face the same PSU/income-conditional loan and performance-contingent merit-scholarship schedule as everyone else, with no unconditional tuition benefit.

We then compare the model-implied difference between the *Gratuidad* baseline and this simulated pre-policy regime to the reduced-form difference-in-differences estimates of subsection 3.2 and subsection 3.3, separately for (i) overall enrollment and enrollment by PSU-eligibility tercile (matching the DiD instrument groups: no public funding, loan-only, and scholarship-eligible), and (ii) intensive-margin outcomes — graduation and dropout — among the cohorts affected by the DiD design. Because the DiD variation is never used to fit the structural parameters, agreement between the two designs is informative about whether the model captures the reform's actual behavioral response rather than an artifact of the post-reform sample it was estimated on.

The model can replicate 2016 effects except for low ability students, but fails short for the 2017 cohort enrollment jump. Predicted enrollment increase in 2017 is arounf 10 p.p. while the model predicts 5 p.p.

Figure 8: Model vs. DiD: Enrollment



Notes:

6 Counterfactuals

We use the estimated model to simulate counterfactual funding policies, evaluating impacts on both enrollment and intensive (effort, credit completion, time-to-degree, graduation) margins. Each policy is defined as a change to the funding rule $OP(x_{it})$ (income and PSU eligibility cutoffs, coverage rate, and performance conditioning), holding the estimated cost primitives $FC_j(\cdot)$, $c_j(\cdot)$, and wages fixed. For each scenario we resolve the full backward induction over the state space and forward-simulate outcomes by Monte Carlo, drawing performance and choice shocks many times per student to average out simulation noise, with effort chosen optimally each period from the FOC in Equation 10. Throughout, the reference scenario is the estimated pre-TFHE regime; each counterfactual is reported as a change relative to this reference.

6.1 Decomposing the effects of each policy

For each counterfactual we decompose the change in outcomes into three channels that are mutually exclusive and exhaustive: a *selection* effect from students whose enrollment decision changes, and, among students who would enroll under either regime, a *price* effect and an *effort* effect.

Let $r \in \{1, ..., R\}$ denote the policy regime and $D_i^r \in \{0, 1\}$ the enrollment decision of student i under regime r. Define the infra-marginal cohort as the students who enroll under the reference (status-quo) regime r',

$$\mathcal{IM} = \{i : D_i^{r'} = 1\},$$

fixed across all counterfactuals so that changes computed on $\mathcal{IM}$ cannot reflect a change in who is enrolled — a policy that excludes some of these students (e.g. through a PSU cutoff) counts them as non-graduates rather than shrinking the denominator. Let Y_{it}^r denote a potential outcome (e.g. graduation) under regime r, written in the structural model as $Y_{it}^r = Y_{it}\big(FC_j^r(x_{it}), c_j^r(x_{it}), e_j^r(x_{it})\big)$. The total effect on the full population decomposes as

$$\underbrace{\mathbb{E}[Y_i^r] - \mathbb{E}[Y_i^{r'}]}_{\Delta Y_{\text{total}}} = \underbrace{\Delta Y_{\text{total}} - \Delta Y_{\text{behav}}}_{\Delta Y_{\text{selection}}} + \underbrace{\Delta Y_{\text{price}}}_{\text{effort fixed}} + \underbrace{\Delta Y_{\text{behav}} - \Delta Y_{\text{price}}}_{\Delta Y_{\text{effort}}},$$

where $\Delta Y_{\text{behav}} = \mathbb{E}[Y_i^r - Y_i^{r'} \mid i \in \mathcal{IM}]$ is the change among infra-marginal students with effort re-optimized, and ΔY_{price} is the same quantity computed with $e_j^r(x_{it})$ mechanically fixed at its pre-policy (status-quo) value, so that any remaining change for $\mathcal{IM}$ can only come from the reduction in out-of-pocket cost. $\Delta Y_{\text{selection}}$ is therefore identified as the residual composition effect of marginal entrants, and ΔY_{effort} isolates the part of the behavioral response that operates through re-optimized studying intensity rather than through relaxed financial constraints — a channel that only the endogenous-effort model can separate from the price channel, since a reduced-form design can at best separate $\Delta Y_{\text{selection}}$ from ΔY_{behav}. We apply this decomposition to each of the scenarios below, and additionally report outcomes split by PSU tercile, SES/income group, and by the latent unobserved type $\tau_i \in \{1, 2\}$ recovered in subsubsection 4.5.1.

6.2 PSU and performance requirements

We saw that TFHE substitutes most of the preexisting aid programs, which were conditional on performance. The removal of performance requirements may have weakened incentives for effort and progression. Preexisting aid programs had performed well in attracting students (Falco & Reichlin, 2025; Solis, 2017, 2026), but the out-of-pocket cost could still be over 20% of tuition. From the Government's perspective, the current policy is very costly: between 2011 and 2019, spending on higher education increased by 160%, reaching 5.4% of GDP, compared with 2.9% in the rest of the OECD (OECD, 2019).

The counterfactuals below build on the conditions of the preexisting aid programs described in subsection 2.1: subsidized loans required $PSU \geq 475$, and merit scholarships required $PSU \geq 500$ and completion of at least 70% of registered credits each year (three of four credit-units in the model's discretized performance measure, $g_{t-1} \geq 3$). They are designed to be budget-improving while preserving the equity target of TFHE. We report four post-reference scenarios: the unrestricted *Gratuidad* reform and three designs that modify the funding rule $OP(x_{it})$ applying specifically to TFHE. The pre-existing loan and scholarship

schedules are left untouched in all four scenarios. The three alternative designs are:

1. **Performance requirement.** Impose the same annual performance requirement used by the pre-existing scholarships ($g_{t-1} \geq 3$) to *retain* free tuition. This restores the performance-contingent component removed by TFHE, without changing who is initially eligible.
   Model prediction: first-period fixed and marginal costs are unaffected, since eligibility is unchanged. From the second year on, students at risk of falling below the threshold face a discrete jump in $OP(x_{it})$ if they fail to comply, so those with the highest marginal and fixed costs of effort should raise effort the most (or drop out if the increase is too costly). If the expected utility of a first year is not compensated by this option value, some students may not enroll at all.

2. **PSU requirement.** Impose $PSU \geq 500$ for *Gratuidad* eligibility, harmonizing the threshold with the pre-existing merit scholarship. This changes the extensive margin only: no performance condition is added, and no fixed or marginal cost changes for students who remain eligible.
   Model prediction: mechanically reduces Government expenditure by shrinking the eligible pool, and attracts, on net, higher-ability students with lower fixed and marginal costs, so persistence and graduation among the enrolled population should improve through selection alone.

3. **Joint requirement.** Combine (1) and (2): $PSU \geq 510$ for eligibility, and the performance requirement to retain funding once enrolled.
   Model prediction: combines the selection and incentive channels above. Reduces the number of beneficiaries the most of the three scenarios, but the remaining recipients face both a more favorable cost distribution and stronger incentives to progress.

Mechanically, (1) affects $OP(x_{it})$ via g_{t-1} for *all* TFHE recipients, tightening incentives within a cohort; (2) affects selection into TFHE at $t = 0$; and (3) compounds both.

Table 4 summarizes the counterfactual results. The three panels distinguish enrollment, graduation, and graduation conditional on enrollment. Entries are percentage-point changes relative to the simulated pre-policy regime, so the columns can be compared directly.

7 Conclusion

This paper investigated the impacts of Tuition-Free Higher Education policies on student enrollment, persistence, and academic performance, using the 2016 Chilean reform as a

Table 4: Counterfactual effects on enrollment, graduation and expected earnings

	Status quo	Change relative to status quo			
	level	Gratuidad	Before-HE performance requirement	During-HE performance requirement	Both requirements
A. Enrollment (%, p.p.)					
All students	60.0	+8.29	-0.12	+7.27	-0.82
Inframarginal students	100.0	0.00	-0.99	0.00	-0.99
Income-eligible students	54.6	+13.43	+0.72	+10.79	-0.42
Low ability	37.5	+11.68	-2.30	+10.96	-2.30
Average ability	55.8	+11.37	+0.22	+9.67	-0.75
High ability	87.1	+1.75	+1.75	+1.12	+0.63
B. Overall graduation (%, p.p.)					
All students	9.9	+1.68	+0.44	+1.41	+0.14
Inframarginal students	17.5	+0.63	+0.24	+0.59	-0.03
Income-eligible students	7.5	+2.85	+0.83	+2.25	+0.34
Low ability	2.6	+1.11	-0.09	+1.26	-0.09
Average ability	6.5	+2.97	+0.45	+2.35	+0.02
High ability	20.6	+0.95	+0.95	+0.63	+0.48
C. Graduation conditional on enrollment (%, p.p.)					
All students	16.4	+0.46	+0.76	+0.32	+0.46
Inframarginal students	17.5	+0.63	+0.42	+0.59	+0.15
Income-eligible students	13.7	+1.47	+1.33	+1.17	+0.74
Low ability	7.0	+0.59	+0.21	+1.01	+0.21
Average ability	11.6	+2.46	+0.76	+1.88	+0.20
High ability	23.7	+0.60	+0.60	+0.41	+0.37
D. Expected lifetime earnings (thousand 2015 USD, %)					
All students	154.0	+5.02	+1.19	+3.61	+0.37
Inframarginal students	184.7	+0.96	+0.39	+0.67	+0.01
Income-eligible students	143.4	+9.02	+2.48	+6.16	+1.03
Low ability	122.3	+5.20	-0.36	+4.24	-0.36
Average ability	141.6	+8.97	+1.33	+6.28	+0.18
High ability	198.4	+2.07	+2.07	+1.30	+0.97

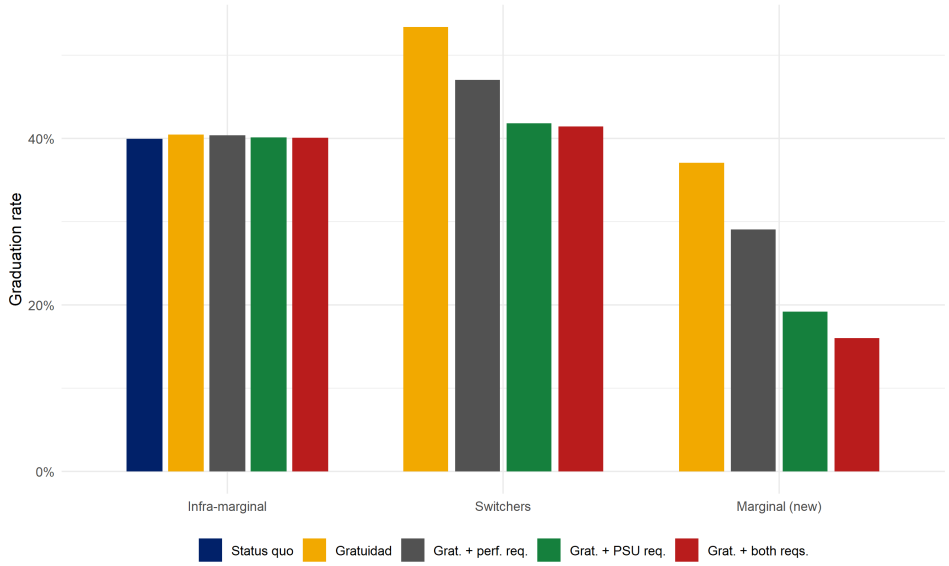
Notes: The first column reports the average level of each outcome for each group under the simulated pre-policy regime (status quo). The remaining columns are changes relative to that level. The units in parentheses in each panel title are those of the status-quo level and of the change: percentage points (p.p.) for enrollment and graduation, percent of the status-quo level for earnings. Gratuidad is unrestricted; the other columns add the named condition. The before-HE performance requirement is the PSU cutoff required to obtain Gratuidad at entry; the during-HE performance requirement is the academic-progress condition for keeping Gratuidad.

Overall graduation uses the full group and counts non-enrollees as non-graduates. Conditional graduation uses the enrollees in each scenario.

Expected lifetime earnings are the present value (5% discount rate, 40 years of experience) of model-predicted earnings, averaged over all students in the group. Non-enrollees earn the no-higher-education wage, graduates the degree wage of their programme, and dropouts the dropout wage, which rises with credits earned. Predictions come from the wage regressions without shocks and measure earnings only: they exclude tuition, effort and other costs, so they are not a welfare measure.

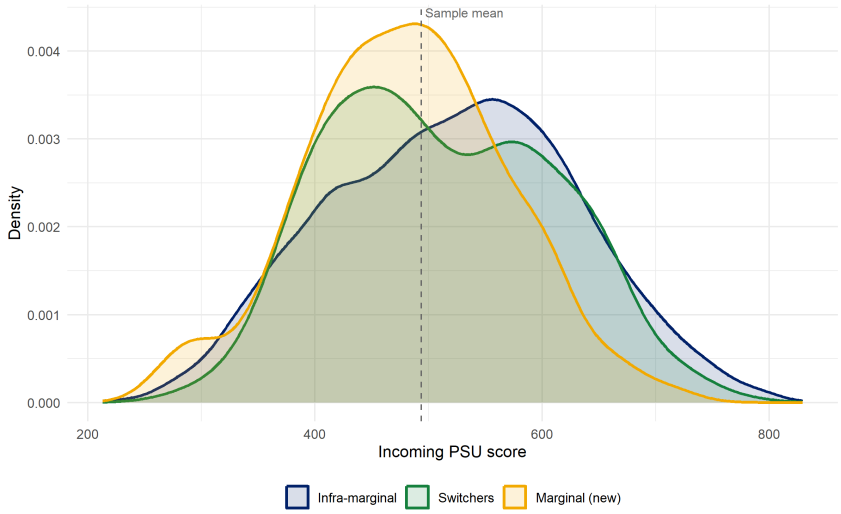
Inframarginal students are those who enroll in the same programme both under the simulated pre-policy regime and under Gratuidad (the same programme, given common random numbers across scenarios); membership is fixed across scenarios, so the change in enrollment in this row is the fraction of them who stop enrolling under a restricted scenario. Students who enroll under both regimes but in a different programme are excluded. Ability is proxied by PSU terciles. Nearly all low-ability students are excluded from Gratuidad eligibility under the before-HE performance requirement, so their near-zero deltas there reflect that exclusion, not a lack of data.

Figure 9: Graduation rate by margin group



Notes: Simulated graduation rate by margin group (infra-marginal students enrolled under both regimes, switchers, and new marginal entrants) and counterfactual scenario, as defined in subsection 6.1.

Figure 10: Incoming ability (PSU) by margin group



Notes: Distribution of incoming PSU score by margin group. The dashed line marks the sample mean.

natural experiment. Our difference-in-differences analysis reveals that the policy significantly boosted enrollment amongst eligible students by 10p.p., driven by lower ability students. Effects on educational outcomes were estimated for infr-marginal students, with a decrease in dropout of 2 p.p. and an increase in on-time graduation of 1.8 p.p.

We built a structural model that endogenizes effort choices, capable of disentangling selection effects from behavioral responses, finding that the removal of performance-based requirements did not lead to reduced effort or weakened outcomes overall. The policy induced a composition effect by attracting student wiht a higher cost of effective effort, while infra-marginal students responded to the policy by increasing effort, as their marginal cost of effort diminished with the reduction in out-of-pocket costs.

These findings have important implications for higher education financing worldwide, suggesting that unconditional aid can promote equity and mobility, especially in contexts with high tuition costs and imperfect income verification.

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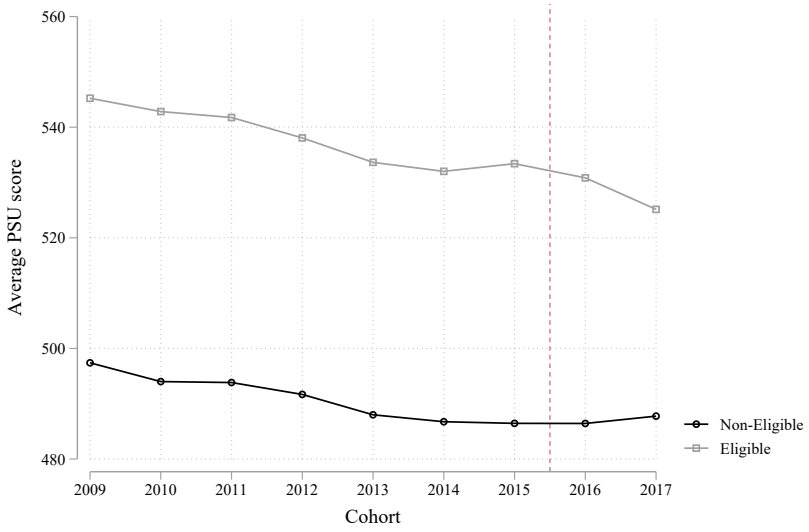
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Appendices

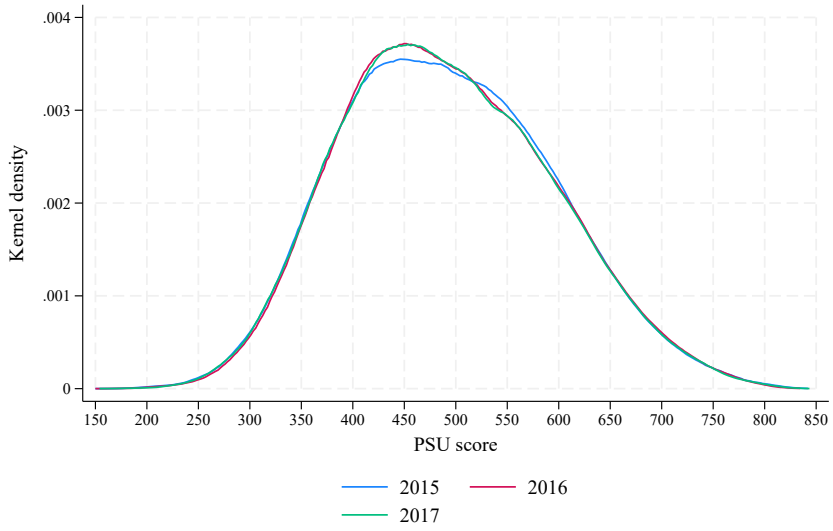
A Figures

Figure A1: Average PSU score over time for participating and non-participating institutions



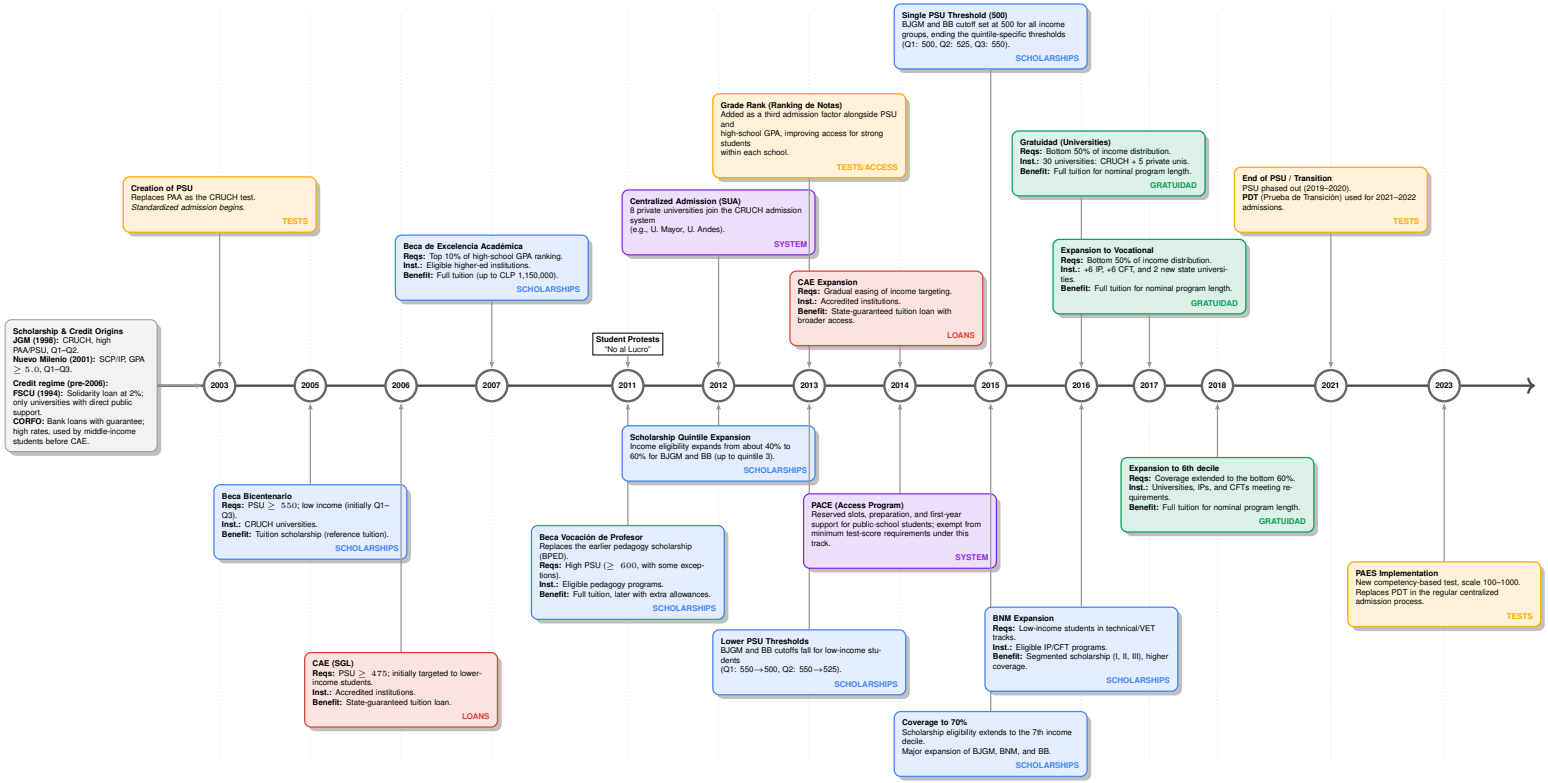
Notes: Annual average PSU score by groups of eligible and non-eligible instutions to free higher education.

Figure A2: Average PSU score over time for participating and non-participating institutions



Notes: Kernel density of PSU average scores enrolled in higher education for years 2015, 2016 and 2017.

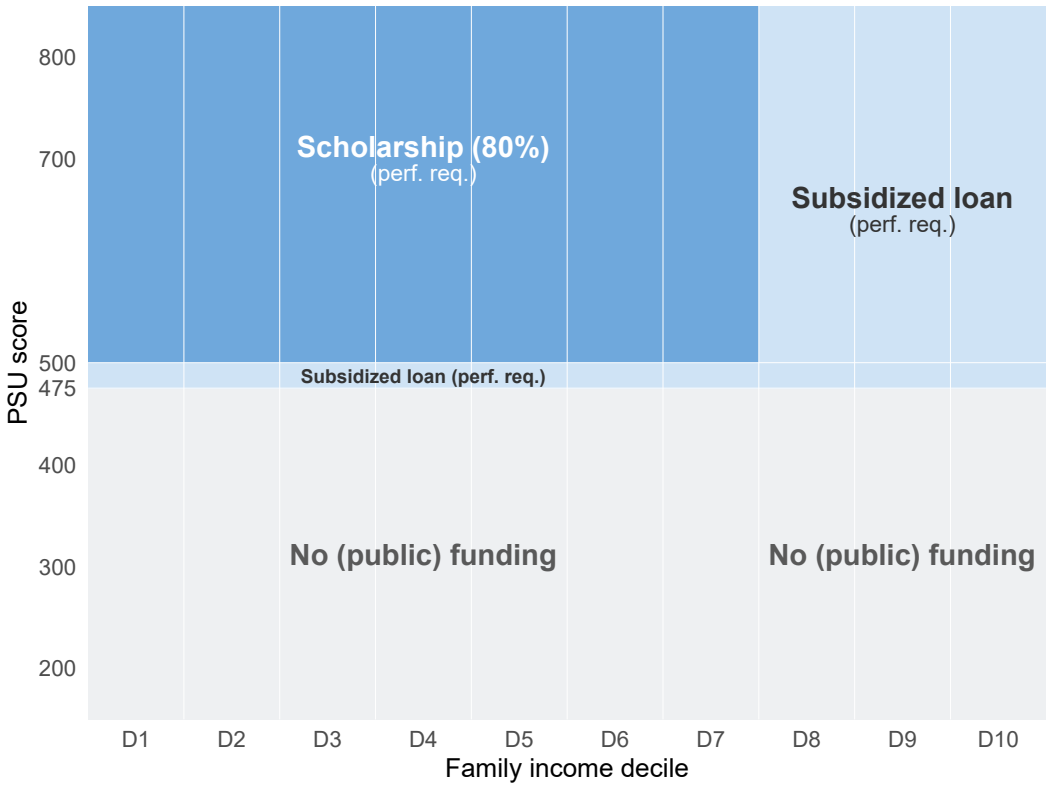
Figure A3: Education reforms 2003-2024



Sources & Notes

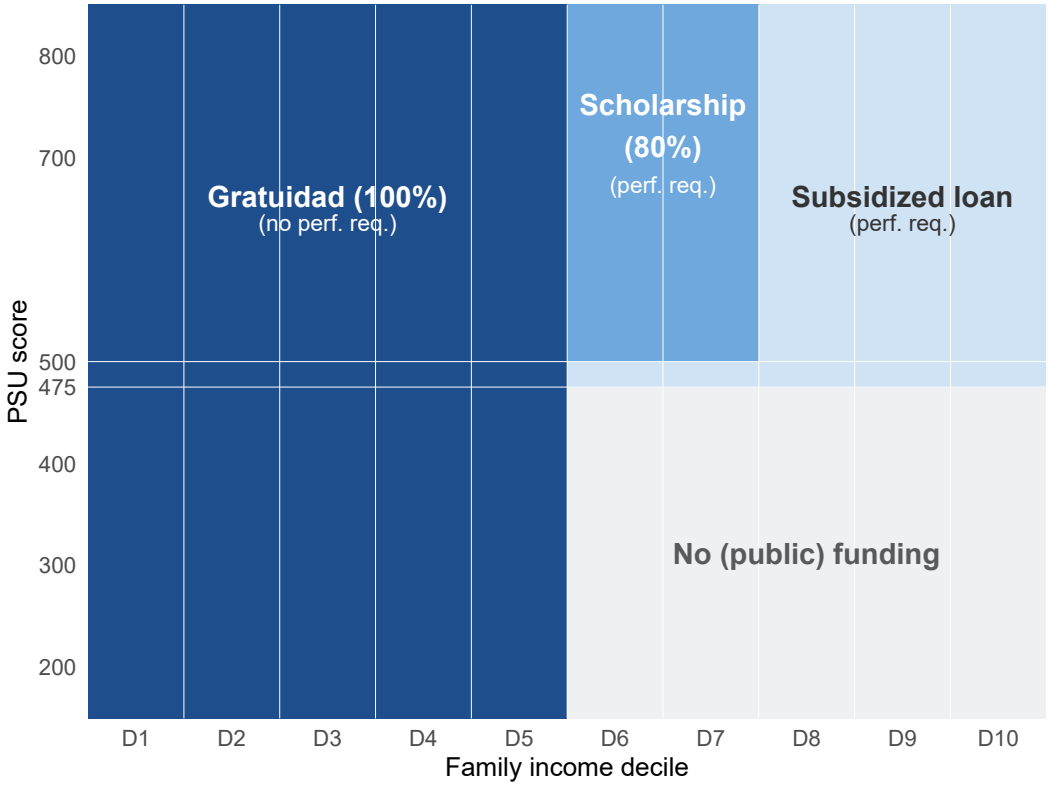
- **Tuition scholarships:** MINEDUC–DIPRES evaluations document the creation and design of Beca Juan Gómez Millas (1998), Beca Nuevo Milenio (2001), Beca Bicentenario (2005), Beca de Excelencia Académica (2007), and the progressive expansion of income coverage.
- **Scholarship and loan regulation:** Decreto 337/2010 and later amendments (Decretos 39/2011, 116/2012, 97/2013), regulating the Higher Education Scholarship Program and its academic and socioeconomic requirements.
- **Admission tests:** DEMRE / Subsecretaría documents on PAA → PSU (2003), Ranking de Notas (2013), PSU → PDT (2021–2022), and PAES (from 2023).
- **CAE:** Law 20.027 and official evaluations describing the initial minimum PSU score, lower-income targeting, and later expansion.
- **FSCU and CORFO:** Law 19.287 and associated rules for FSCU, plus CORFO student-loan rules, defining the pre-CAE funding regime.
- **Gratuidad 2016 (universities):** Initial network = 30 universities. Private universities in the 2016 cohort were: UAH, UCSH, UDP, UAutónoma, and UFT.
- **Gratuidad 2017 (SCP/IP expansion):** The policy expanded to 44 institutions in total, adding 6 IP, 6 CFT, and the two new state universities.
- **Later additions to the Gratuidad network:** Universidad Santo Tomás joined in 2022; Universidad Central joined in 2023.
- **PACE:** Exenta 680/2015 formally establishes PACE and its access/support design.
- **2015 scholarship thresholds and coverage:** Mineduc, *Balance de Gestión Integral 2015* (Subsecretaría de Educación, April 2016), section 3 (Becas y Créditos): PSU requirement for BB and BJGM lowered to 500 points and coverage extended to the 7th income decile from 2015.
- **SUA and private universities:** CRUCH/DEMRE documentation and press sources describe the incorporation of eight private universities into the centralized admission system for the 2012 process.

Figure A4: Funding instruments by PSU score and family income, 2015



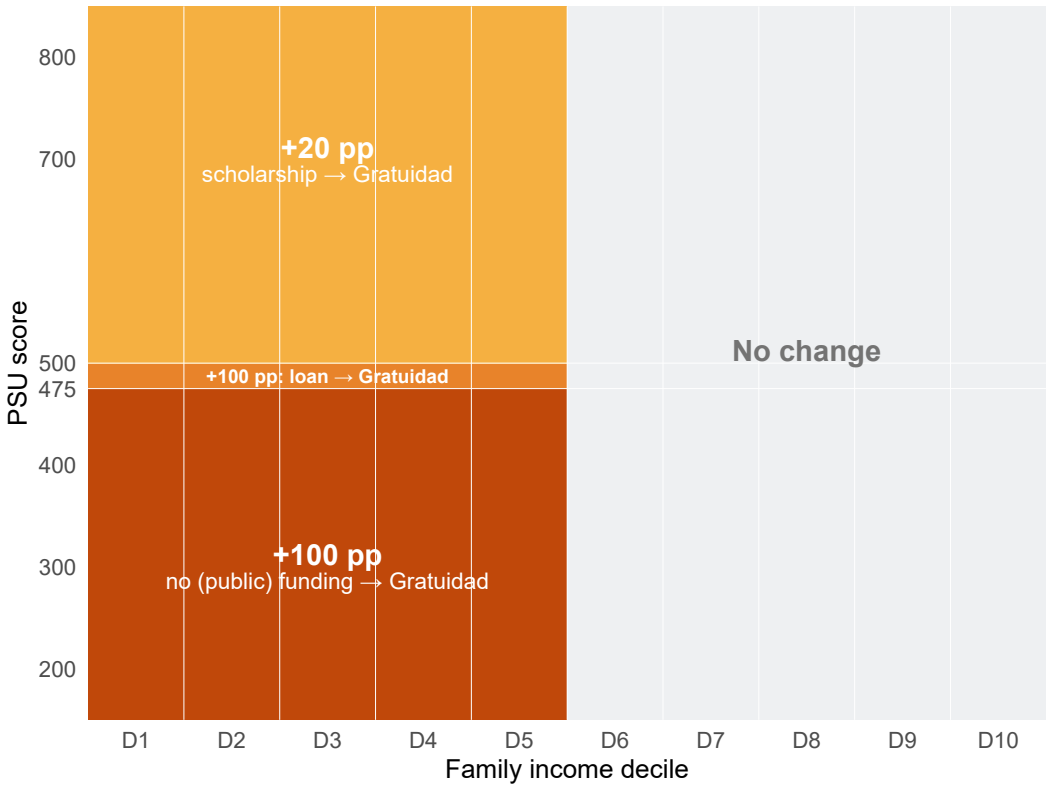
Notes: Funding instrument available to students by PSU score (vertical axis) and family income decile (horizontal axis) before the TFHE policy. Scholarships finance 80% of reference tuition and loans are subsidized; all instruments require meeting academic performance requirements.

Figure A5: Funding instruments by PSU score and family income, 2016



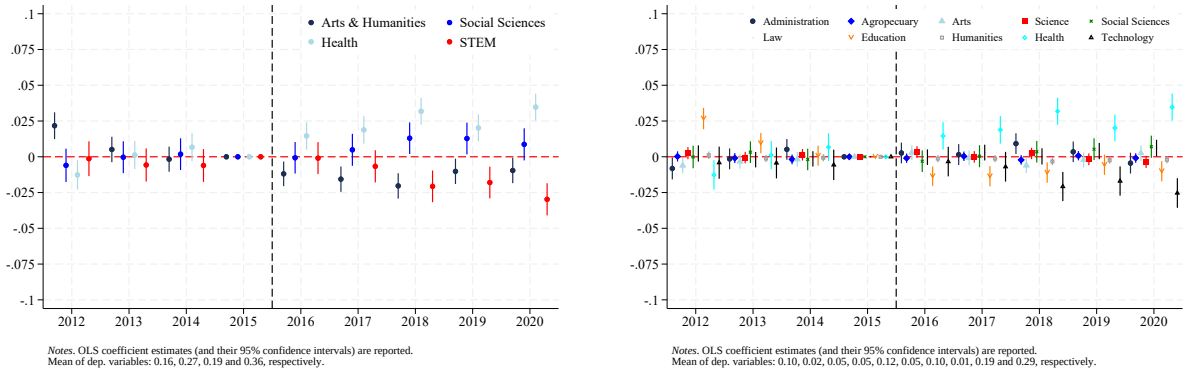
Notes: Funding instrument available to students by PSU score (vertical axis) and family income decile (horizontal axis) after the introduction of TFHE (*Gratuidad*) for the 2016 cohort. *Gratuidad* covers 100% of reference tuition with no performance requirement and applies to students in income deciles 1 to 5 attending eligible institutions.

Figure A6: Change in funding instruments after TFHE



Notes: Change in the share of students receiving *Gratuidad* between 2015 and 2016, by PSU score (vertical axis) and family income decile (horizontal axis), and the instrument it substitutes for. No change for deciles 6 to 10.

Figure A7: DID: Preferences



a) 4 fields

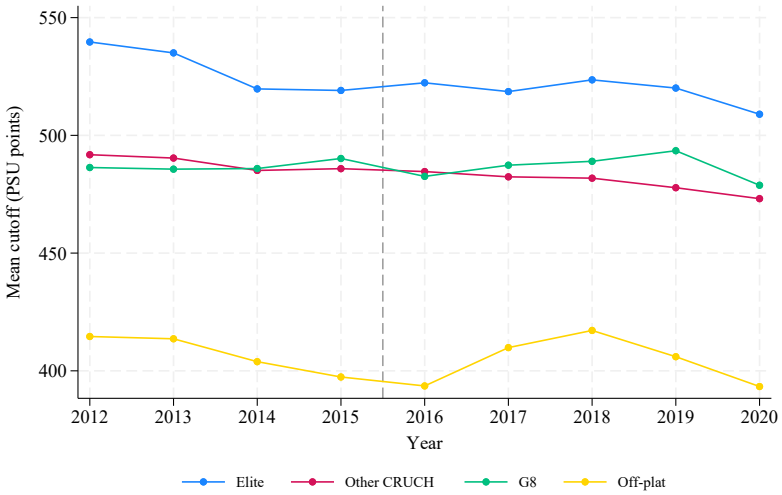
b) 10 fields

Notes: These figures show the results of estimating Equation 1. Panel (a) groups the results in four fields, while panel (b) uses the original ten-field classification. STEM includes Agropecuary, Science and Technology. Social Sciences includes Administration, Social Sciences nad Law. Arts and Humanities includes Education, Arts and Humanities. Health includes Health Sciences.

A.1 Descriptive evidence on spillovers (SUTVA)

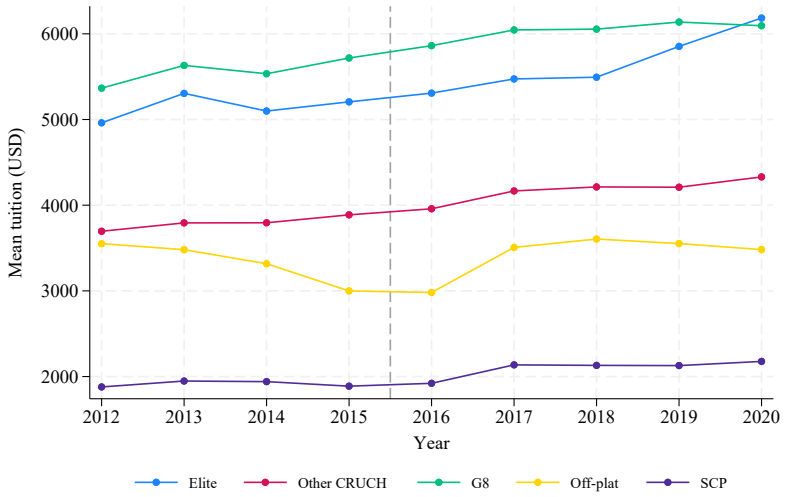
The DiD design in section 3 requires that the policy does not affect non-eligible students through changes in institutions' supply, selectivity or prices. Figure A8–A10 summarize the supply-side margins through which such spillovers could operate: admission cutoffs, prices, and vacancies.

Figure A8: Mean admission cutoff by institution tier



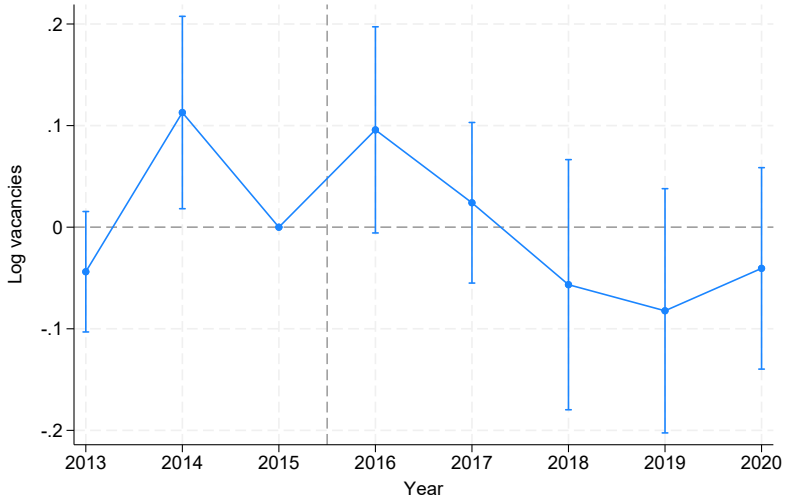
Notes: Mean PSU cutoff of bachelor programs, by tier. Elite: Pontificia Universidad Católica de Chile, Universidad de Chile, Universidad de Concepción, Universidad Técnica Federico Santa María, Pontificia Universidad Católica de Valparaíso and Universidad de Santiago de Chile. G8: Universidad Adolfo Ibáñez, Alberto Hurtado, Andrés Bello, de los Andes, del Desarrollo, Diego Portales, Finis Terrae and Mayor. Other CRUCH: the remaining universities of the Council of Rectors (CRUCH). Off-plat: universities outside CRUCH that are not in the G8. The dashed line marks the introduction of TFHE.

Figure A9: Mean tuition by institution tier



Notes: Mean annual tuition (USD) of programs, by tier. Tiers as in Figure A8: Elite (PUC, U. de Chile, U. de Concepción, UTFSM, PUCV, USACH); G8 (UAI, UAH, UNAB, U. de los Andes, UDD, UDP, U. Finis Terrae, U. Mayor); Other CRUCH (remaining CRUCH universities); Off-plat (non-CRUCH, non-G8 universities); SCP (short-cycle programs at any institution). The dashed line marks the introduction of TFHE.

Figure A10: DiD event study: log vacancies of bachelor programs



Notes: Coefficients of an event study of log vacancies on the interaction between year and an indicator for programs at institutions adhering to TFHE (relative to non-adhering universities), with program and year fixed effects and 2015 as the omitted year. Standard errors clustered by institution; bars show 95% confidence intervals. Only a few non-adhering universities have a 2015 cutoff, so inference rests on few control clusters.

B Tables

Table B1: Enrollment and outcomes, quasi-experimental sample

Extensive margin	2013	2014	2015	2016	2017	2018	2019	2020
A. Enrollment								
Enrolled in higher education	0.626	0.633	0.631	0.640	0.658	0.659	0.646	0.662
University enrollment	0.308	0.314	0.313	0.343	0.337	0.343	0.338	0.369
SCP enrollment	0.316	0.319	0.318	0.298	0.321	0.316	0.308	0.293
Not enrolled	0.374	0.367	0.369	0.360	0.342	0.341	0.354	0.338
B. Controls								
Female	0.553	0.551	0.551	0.548	0.553	0.549	0.544	0.547
Family income bracket (1–10)	3.33	3.59	3.75	3.95	4.16	4.04	4.06	4.40
PSU	486.7	487.8	490.3	495.2	495.8	499.9	494.4	497.6
GPA	541.2	542.1	544.0	550.5	555.1	559.0	563.6	576.7
C. Funding								
Gratuidad	0.000	0.000	0.000	0.339	0.558	0.596	0.585	0.587
Subsidized loan	0.308	0.310	0.292	0.188	0.120	0.110	0.117	0.108
Merit scholarship	0.390	0.388	0.442	0.247	0.118	0.111	0.121	0.120
No (public) funding	0.302	0.302	0.266	0.226	0.204	0.182	0.177	0.186
Observations	102,480	104,348	106,956	108,695	103,951	119,227	127,146	118,717

Intensive margin	2009	2010	2011	2012	2013	2014	2015	2016	2017
D. Outcomes									
Graduation	0.358	0.353	0.344	0.364	0.366	0.392	0.385	0.379	0.365
On-time graduation	0.168	0.151	0.154	0.169	0.177	0.199	0.188	0.179	0.206
Persistence (years enrolled)	3.51	3.42	3.36	3.39	3.33	3.42	3.42	3.50	3.51
Field switch	0.267	0.270	0.268	0.264	0.263	0.246	0.235	0.217	0.195
Dropout	0.160	0.168	0.164	0.157	0.170	0.182	0.196	0.226	0.269
Observations	51,872	56,332	58,333	55,962	64,015	66,059	67,490	69,609	68,391

Notes: The extensive sample follows the pooled enrollment DiD; the intensive sample follows the pooled enrolled-student DiD. University/SCP rows are unconditional extensive-sample shares. Funding shares condition on enrollment and use the priority Gratuidad, loan, then merit scholarship. Characteristics are the reduced-form controls (family income defines the control group). Observations are students in the extensive sample (Panels A–C) and enrolled students in the intensive sample (Panel D). On-time graduation means within nominal duration.

Table B2: Descriptive statistics, model sample

	2016	2017
A. Students and entry choices		
University enrollment	0.359	0.284
SCP enrollment	0.349	0.316
Not enrolled	0.292	0.401
B. Student characteristics		
Female	0.512	0.527
Family income bracket (1–10)	4.02	4.45
PSU	493.7	492.3
GPA	549.6	547.5
C. Funding		
Gratuidad	0.253	0.414
Subsidized loan	0.222	0.160
Merit scholarship	0.267	0.177
None of these funding sources	0.259	0.250
D. Academic progress		
Registered / annual curriculum	0.935	0.932
Completed / annual curriculum	0.810	0.815
Completed / registered	0.844	0.854
Observations (student-years)	15,470	13,162
Observations (students)	6,981	6,993

Notes: Cohorts 2016–2017 in the current model-estimation input. Panels A–C count each student once (Observations (students)) at entry, including the outside option; funding conditions on entry enrollment and uses the priority Gratuidad, loan, then merit scholarship. PSU and GPA are original scores, not standardized model regressors. Panel D weights each chosen enrolled student-year equally over the observed model horizon. Credit rates are normalized by the annual curriculum, and pool credit- and course-reporting programs.

	dropout	
OP fees	0.101***	(0.006)
Rel. Ability	-0.105***	(0.023)
Diff. Region	0.106***	(0.026)
Credits left	-0.113***	(0.004)
Designed total credits	0.099***	(0.006)
Delay	1.098***	(0.051)
Performance: category 0	2.116***	(0.040)
Performance: category 1	1.435***	(0.040)
Performance: category 2	0.291***	(0.033)
Performance: category 4	-0.721***	(0.049)
female	0.197**	(0.079)
Middle SES	0.033	(0.098)
High SES	0.007	(0.100)
PSU	0.063	(0.066)
GPA	-0.111**	(0.053)
Type 2	-10.276***	(0.656)
OP × female	-0.010**	(0.005)
OP × middle SES	-0.030***	(0.007)
OP × high SES	-0.027***	(0.008)
OP × psu	-0.026***	(0.004)
OP × gpa	0.015***	(0.003)
Delay × female	0.057	(0.050)
Delay × middle SES	0.040	(0.062)
Delay × high SES	0.233***	(0.062)
Delay × PSU	-0.191***	(0.035)
Delay × GPA	-0.062*	(0.034)
Delay × SCP	0.584***	(0.042)
t=2 × SCP	0.446***	(0.030)
t=3 × SCP	0.832***	(0.047)
t=4 × SCP	0.988***	(0.087)
t=5 × SCP	0.830***	(0.184)
t=6 × SCP	0.910**	(0.391)
Performance 0 × female	-0.301***	(0.039)
Performance 0 × middle SES	-0.060	(0.046)
Performance 0 × high SES	-0.093*	(0.051)
Performance 0 × PSU	-0.022	(0.027)
Performance 0 × GPA	-0.011	(0.026)
Performance 1 × female	-0.278***	(0.041)
Performance 1 × middle SES	-0.024	(0.048)
Performance 1 × high SES	0.053	(0.053)
Performance 1 × PSU	0.005	(0.027)
Performance 1 × GPA	0.072***	(0.027)
Performance 2 × female	-0.099***	(0.035)
Performance 2 × middle SES	-0.028	(0.043)
Performance 2 × high SES	0.082*	(0.046)
Performance 2 × PSU	-0.103***	(0.024)

Performance 2 × GPA	0.079***	(0.024)
Performance 4 × female	-0.328***	(0.052)
Performance 4 × middle SES	-0.046	(0.065)
Performance 4 × high SES	0.263***	(0.065)
Performance 4 × PSU	-0.180***	(0.036)
Performance 4 × GPA	-0.091**	(0.035)
Credits left × female	0.001	(0.002)
Credits left × middle SES	0.001	(0.003)
Credits left × high SES	-0.007**	(0.003)
Credits left × psu	0.007***	(0.002)
Credits left × gpa	0.010***	(0.002)
Rel. Abil. × female	0.038***	(0.011)
Rel. Abil. × middle SES	0.011	(0.013)
Rel. Abil. × high SES	0.057***	(0.015)
Rel. Abil. × psu	0.011*	(0.006)
Rel. Abil. × gpa	0.029***	(0.007)
Diff. Region × female	0.009	(0.027)
Diff. Region × middle SES	0.021	(0.034)
Diff. Region × high SES	0.037	(0.034)
Diff. Region × psu	0.034*	(0.018)
Diff. Region × gpa	-0.055***	(0.018)
Agriculture × fem	-0.073	(0.089)
Agriculture × middle SES	-0.034	(0.110)
Agriculture × high SES	-0.403***	(0.107)
Agriculture × psu	0.094	(0.069)
Agriculture × gpa	0.087	(0.059)
Agriculture × Type 2	0.055	(0.104)
Arts × fem	0.167**	(0.067)
Arts × middle SES	-0.011	(0.086)
Arts × high SES	-0.114	(0.080)
Arts × psu	0.077	(0.053)
Arts × gpa	-0.005	(0.046)
Arts × Type 2	-0.069	(0.084)
Science × fem	-0.047	(0.063)
Science × middle SES	-0.122	(0.081)
Science × high SES	0.064	(0.076)
Science × psu	-0.067	(0.050)
Science × gpa	0.226***	(0.043)
Science × Type 2	-1.016***	(0.084)
Social Sciences × fem	0.009	(0.055)
Social Sciences × middle SES	-0.039	(0.070)
Social Sciences × high SES	0.062	(0.066)
Social Sciences × psu	0.243***	(0.042)
Social Sciences × gpa	0.112***	(0.037)
Social Sciences × Type 2	0.014	(0.072)
Law × fem	-0.068	(0.060)
Law × middle SES	0.023	(0.076)

Law × high SES	-0.109	(0.071)
Law × psu	0.161***	(0.047)
Law × gpa	0.025	(0.040)
Law × Type 2	-0.536***	(0.074)
Education × fem	0.212***	(0.057)
Education × middle SES	-0.020	(0.067)
Education × high SES	-0.010	(0.071)
Education × psu	0.038	(0.041)
Education × gpa	0.076**	(0.037)
Education × Type 2	-0.408***	(0.069)
Humanities × fem	-0.036	(0.132)
Humanities × middle SES	-0.191	(0.141)
Humanities × high SES	-0.059	(0.152)
Humanities × psu	0.194**	(0.089)
Humanities × gpa	0.067	(0.081)
Humanities × Type 2	-0.451***	(0.170)
SCP × fem	0.053	(0.092)
SCP × middle SES	-0.154	(0.108)
SCP × high SES	-0.231*	(0.134)
SCP × psu	0.321***	(0.067)
SCP × gpa	0.064	(0.059)
SCP × type 2	1.215***	(0.128)
SCP-Gratuidad × fem	0.263***	(0.049)
SCP-Gratuidad × middle SES	-0.020	(0.060)
SCP-Gratuidad × high SES	-0.243***	(0.064)
SCP-Gratuidad × psu	0.408***	(0.039)
SCP-Gratuidad × gpa	0.215***	(0.033)
SCP-Gratuidad × type 2	0.796***	(0.073)
Health × fem	-0.024	(0.053)
Health × middle SES	-0.025	(0.064)
Health × high SES	-0.146**	(0.063)
Health × psu	0.133***	(0.040)
Health × gpa	-0.074**	(0.034)
Health × type 2	0.291***	(0.065)
Technology × fem	-0.078	(0.048)
Technology × middle SES	-0.026	(0.059)
Technology × high SES	-0.194***	(0.057)
Technology × psu	0.318***	(0.036)
Technology × gpa	0.141***	(0.031)
Technology × type 2	-0.417***	(0.057)
OP × Type 2	0.034***	(0.006)
Delay × Type 2	4.829***	(0.621)
Performance 0 × Type 2	4.638***	(0.191)
Performance 1 × Type 2	3.665***	(0.191)
Performance 2 × Type 2	3.259***	(0.190)
Performance 4 × Type 2	3.594***	(0.295)
Credits left × Type 2	0.170***	(0.005)

Rel. Abil. x Type 2	0.042***	(0.014)
Dist. x Type 2	-0.022	(0.034)
t= 2	-0.882***	(0.021)
t= 3	-1.523***	(0.030)
t= 4	-1.734***	(0.041)
t= 5	-1.839***	(0.053)
t= 6	-1.782***	(0.062)
Agricultural	-0.588***	(0.092)
Arts and Architecture	-0.026	(0.072)
Science	0.801***	(0.070)
Social Sciences	-0.010	(0.058)
Law	-0.100	(0.065)
Education	0.252***	(0.057)
Humanities	0.674***	(0.135)
SCP	0.933***	(0.134)
SCP-Gratuidad	0.440***	(0.105)
Health	-0.531***	(0.057)
Technology	-0.196***	(0.048)
mineduc=3	-0.233***	(0.050)
mineduc=10	-0.276***	(0.054)
mineduc=20	0.897***	(0.063)
mineduc=31	-0.038	(0.063)
mineduc=39	-0.168***	(0.058)
mineduc=45	-0.259***	(0.052)
mineduc=54	0.415***	(0.081)
mineduc=71	-0.084*	(0.048)
mineduc=72	-0.311***	(0.052)
mineduc=75	-0.347***	(0.059)
mineduc=76	-0.393***	(0.054)
mineduc=78	-0.528***	(0.055)
mineduc=85	-0.289***	(0.062)
mineduc=86	-0.407***	(0.036)
mineduc=87	-0.584***	(0.042)
mineduc=88	-0.348***	(0.042)
mineduc=89	-0.711***	(0.044)
mineduc=90	-0.886***	(0.050)
mineduc=93	-0.909***	(0.068)
mineduc=94	-0.303***	(0.062)
mineduc=990	-0.168***	(0.052)
mineduc=991	0.169***	(0.048)
mineduc=992	-0.410***	(0.050)
mineduc=993	-0.448***	(0.021)
Observations	457,037	
Mean of Dep. Variable	0.205	

Standard errors in parentheses

Logistic regression of choosing the outside option (dropping out) in period t. Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardised. Base category is low SES, mean ability, enrolled in Business at Universidad de Chile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B4: Performance ordered logit

	index	
female	-0.052	(0.052)
Middle SES	0.016	(0.062)
High SES	-0.180***	(0.065)
PSU	0.116***	(0.044)
GPA	0.173***	(0.035)
Type 2	-3.534***	(0.102)
OP fees	-0.042***	(0.004)
Rel. Ability	0.038***	(0.014)
Diff. Region	-0.039**	(0.017)
Credits left (t-1)	0.152***	(0.004)
Designed total credits	-0.170***	(0.004)
Delay (t-1)	-1.694***	(0.029)
Performance (t-1): category 0	-1.128***	(0.040)
Performance (t-1): category 1	-0.839***	(0.041)
Performance (t-1): category 2	-0.257***	(0.028)
Performance (t-1): category 4	0.846***	(0.033)
OP × female	0.008***	(0.003)
OP × middle SES	0.031***	(0.004)
OP × high SES	0.037***	(0.005)
OP × psu	-0.006**	(0.002)
OP × gpa	0.004**	(0.002)
Delay (t-1) × female	-0.165***	(0.025)
Delay (t-1) × middle SES	-0.003	(0.030)
Delay (t-1) × high SES	0.021	(0.031)
Delay (t-1) × psu	0.156***	(0.019)
Delay (t-1) × gpa	0.008	(0.017)
Delay (t-1) × scp	0.455***	(0.033)
t = 2 × SCP	-0.489***	(0.026)
t = 3 × SCP	-1.680***	(0.042)
t = 4 × SCP	-3.042***	(0.071)
t = 5 × SCP	-3.302***	(0.146)
t = 6 × SCP	-2.925***	(0.325)
t = 7 × SCP	-1.733***	(0.670)
Performance (t-1) 0 × female	-0.101***	(0.034)
Performance (t-1) 0 × middle SES	-0.002	(0.041)
Performance (t-1) 0 × high SES	-0.069	(0.043)
Performance (t-1) 0 × PSU	0.077***	(0.025)
Performance (t-1) 0 × GPA	0.108***	(0.023)
Performance (t-1) 1 × female	0.172***	(0.043)
Performance (t-1) 1 × middle SES	0.061	(0.050)
Performance (t-1) 1 × high SES	0.094*	(0.054)
Performance (t-1) 1 × PSU	-0.281***	(0.029)
Performance (t-1) 1 × GPA	0.043	(0.028)
Performance (t-1) 2 × female	0.116***	(0.030)
Performance (t-1) 2 × middle SES	-0.009	(0.036)
Performance (t-1) 2 × high SES	0.055	(0.038)
Performance (t-1) 2 × PSU	-0.150***	(0.021)
Performance (t-1) 2 × GPA	0.089***	(0.020)
Performance (t-1) 4 × female	0.043	(0.034)
Performance (t-1) 4 × middle SES	-0.032	(0.040)
Performance (t-1) 4 × high SES	0.026	(0.042)
Performance (t-1) 4 × PSU	-0.069***	(0.024)
Performance (t-1) 4 × GPA	0.130***	(0.022)
Credits left (t-1) × female	0.025***	(0.002)
Credits left (t-1) × middle SES	0.005**	(0.002)
Credits left (t-1) × high SES	0.012***	(0.002)
Credits left (t-1) × psu	0.010***	(0.001)
Credits left (t-1) × gpa	0.006***	(0.001)
Rel. Abil. × female	-0.004	(0.007)
Rel. Abil. × middle SES	-0.001	(0.008)
Rel. Abil. × high SES	0.003	(0.010)
Rel. Abil. × psu	-0.019***	(0.004)
Rel. Abil. × gpa	0.035***	(0.004)
Diff. Region × female	-0.034**	(0.017)
Diff. Region × middle SES	0.004	(0.021)
Diff. Region × high SES	-0.043**	(0.021)
Diff. Region × psu	-0.007	(0.012)
Diff. Region × gpa	0.019*	(0.012)
OP × Type 2	-0.009**	(0.004)
Delay (t-1) × Type 2	1.069***	(0.048)
Performance (t-1) 0 × Type 2	-0.249***	(0.067)
Performance (t-1) 1 × Type 2	-0.062	(0.064)
Performance (t-1) 2 × Type 2	-0.207***	(0.056)
Performance (t-1) 4 × Type 2	-0.336***	(0.074)
Credits left (t-1) × Type 2	0.009**	(0.004)
Rel. Abil. x Type 2	0.044***	(0.009)
Dist. x Type 2	-0.090***	(0.023)
t = 2	-1.031***	(0.030)
t = 3	-0.079**	(0.035)
t = 4	0.999***	(0.041)
t = 5	1.558***	(0.049)
t = 6	1.100***	(0.057)
t = 7	0.390***	(0.063)
Observations	553,563	
Mean of Dep. Variable	3.113	

	index	
female	-0.052	(0.052)
Middle SES	0.016	(0.062)
High SES	-0.180***	(0.065)
PSU	0.116***	(0.044)
GPA	0.173***	(0.035)
Type 2	-3.534***	(0.102)
OP fees	-0.042***	(0.004)
Rel. Ability	0.038***	(0.014)
Diff. Region	-0.039**	(0.017)
Credits left (t-1)	0.152***	(0.004)
Designed total credits	-0.170***	(0.004)
Delay (t-1)	-1.694***	(0.029)
Performance (t-1): category 0	-1.128***	(0.040)
Performance (t-1): category 1	-0.839***	(0.041)
Performance (t-1): category 2	-0.257***	(0.028)
Performance (t-1): category 4	0.846***	(0.033)
OP × female	0.008***	(0.003)
OP × middle SES	0.031***	(0.004)
OP × high SES	0.037***	(0.005)
OP × psu	-0.006**	(0.002)
OP × gpa	0.004**	(0.002)
Delay (t-1) × female	-0.165***	(0.025)
Delay (t-1) × middle SES	-0.003	(0.030)
Delay (t-1) × high SES	0.021	(0.031)
Delay (t-1) × psu	0.156***	(0.019)
Delay (t-1) × gpa	0.008	(0.017)
Delay (t-1) × scp	0.455***	(0.033)
t = 2 × SCP	-0.489***	(0.026)
t = 3 × SCP	-1.680***	(0.042)
t = 4 × SCP	-3.042***	(0.071)
t = 5 × SCP	-3.302***	(0.146)
t = 6 × SCP	-2.925***	(0.325)
t = 7 × SCP	-1.733***	(0.670)
Performance (t-1) 0 × female	-0.101***	(0.034)
Performance (t-1) 0 × middle SES	-0.002	(0.041)
Performance (t-1) 0 × high SES	-0.069	(0.043)
Performance (t-1) 0 × PSU	0.077***	(0.025)
Performance (t-1) 0 × GPA	0.108***	(0.023)
Performance (t-1) 1 × female	0.172***	(0.043)
Performance (t-1) 1 × middle SES	0.061	(0.050)
Performance (t-1) 1 × high SES	0.094*	(0.054)
Performance (t-1) 1 × PSU	-0.281***	(0.029)
Performance (t-1) 1 × GPA	0.043	(0.028)
Performance (t-1) 2 × female	0.116***	(0.030)
Performance (t-1) 2 × middle SES	-0.009	(0.036)
Performance (t-1) 2 × high SES	0.055	(0.038)

Performance (t-1) 2 × PSU	-0.150***	(0.021)
Performance (t-1) 2 × GPA	0.089***	(0.020)
Performance (t-1) 4 × female	0.043	(0.034)
Performance (t-1) 4 × middle SES	-0.032	(0.040)
Performance (t-1) 4 × high SES	0.026	(0.042)
Performance (t-1) 4 × PSU	-0.069***	(0.024)
Performance (t-1) 4 × GPA	0.130***	(0.022)
Credits left (t-1) × female	0.025***	(0.002)
Credits left (t-1) × middle SES	0.005**	(0.002)
Credits left (t-1) × high SES	0.012***	(0.002)
Credits left (t-1) × psu	0.010***	(0.001)
Credits left (t-1) × gpa	0.006***	(0.001)
Rel. Abil. × female	-0.004	(0.007)
Rel. Abil. × middle SES	-0.001	(0.008)
Rel. Abil. × high SES	0.003	(0.010)
Rel. Abil. × psu	-0.019***	(0.004)
Rel. Abil. × gpa	0.035***	(0.004)
Diff. Region × female	-0.034**	(0.017)
Diff. Region × middle SES	0.004	(0.021)
Diff. Region × high SES	-0.043**	(0.021)
Diff. Region × psu	-0.007	(0.012)
Diff. Region × gpa	0.019*	(0.012)
Agriculture × fem	-0.246***	(0.048)
Agriculture × middle SES	-0.009	(0.059)
Agriculture × high SES	0.016	(0.057)
Agriculture × psu	-0.193***	(0.039)
Agriculture × gpa	-0.126***	(0.033)
Agriculture × Type 2	0.306***	(0.069)
Arts × fem	-0.021	(0.042)
Arts × middle SES	-0.024	(0.054)
Arts × high SES	0.139***	(0.051)
Arts × psu	-0.140***	(0.035)
Arts × gpa	-0.153***	(0.029)
Arts × Type 2	-0.360***	(0.056)
Science × fem	-0.241***	(0.042)
Science × middle SES	0.019	(0.053)
Science × high SES	0.032	(0.051)
Science × psu	-0.038	(0.034)
Science × gpa	-0.205***	(0.029)
Science × Type 2	0.251***	(0.056)
Social Sciences × fem	0.055	(0.036)
Social Sciences × middle SES	-0.026	(0.044)
Social Sciences × high SES	-0.059	(0.043)
Social Sciences × psu	-0.186***	(0.027)
Social Sciences × gpa	-0.106***	(0.024)
Social Sciences × Type 2	-0.316***	(0.047)
Law × fem	-0.207***	(0.037)

Law × middle SES	-0.074	(0.046)
Law × high SES	0.103**	(0.045)
Law × psu	-0.072**	(0.029)
Law × gpa	-0.107***	(0.025)
Law × Type 2	0.220***	(0.051)
Education × fem	0.075**	(0.037)
Education × middle SES	-0.005	(0.043)
Education × high SES	0.116**	(0.048)
Education × psu	-0.111***	(0.028)
Education × gpa	-0.099***	(0.024)
Education × Type 2	-0.258***	(0.047)
Humanities × fem	-0.006	(0.096)
Humanities × middle SES	-0.048	(0.099)
Humanities × high SES	0.097	(0.111)
Humanities × psu	-0.047	(0.065)
Humanities × gpa	-0.184***	(0.059)
Humanities × Type 2	0.104	(0.114)
SCP × fem	0.209***	(0.063)
SCP × middle SES	-0.030	(0.073)
SCP × high SES	0.297***	(0.095)
SCP × psu	0.058	(0.046)
SCP × gpa	-0.067	(0.041)
SCP × type 2	-0.625***	(0.077)
SCP-Gratuidad × fem	0.283***	(0.034)
SCP-Gratuidad × middle SES	0.057	(0.040)
SCP-Gratuidad × high SES	0.184***	(0.045)
SCP-Gratuidad × psu	0.072***	(0.027)
SCP-Gratuidad × gpa	0.055**	(0.023)
SCP-Gratuidad × type 2	-0.356***	(0.055)
Health × fem	-0.066**	(0.031)
Health × middle SES	-0.042	(0.037)
Health × high SES	0.113***	(0.037)
Health × psu	-0.000	(0.024)
Health × gpa	-0.117***	(0.020)
Health × type 2	-0.048	(0.043)
Technology × fem	-0.136***	(0.030)
Technology × middle SES	-0.039	(0.035)
Technology × high SES	0.038	(0.035)
Technology × psu	0.007	(0.023)
Technology × gpa	-0.061***	(0.019)
Technology × type 2	0.205***	(0.039)
OP × Type 2	-0.009**	(0.004)
Delay (t-1) × Type 2	1.069***	(0.048)
Performance (t-1) 0 × Type 2	-0.249***	(0.067)
Performance (t-1) 1 × Type 2	-0.062	(0.064)
Performance (t-1) 2 × Type 2	-0.207***	(0.056)
Performance (t-1) 4 × Type 2	-0.336***	(0.074)

Credits left (t-1) × Type 2	0.009**	(0.004)
Rel. Abil. x Type 2	0.044***	(0.009)
Dist. x Type 2	-0.090***	(0.023)
t = 2	-1.031***	(0.030)
t = 3	-0.079**	(0.035)
t = 4	0.999***	(0.041)
t = 5	1.558***	(0.049)
t = 6	1.100***	(0.057)
t = 7	0.390***	(0.063)
Agricultural	-0.077	(0.049)
Arts and Architecture	0.074*	(0.045)
Science	-0.606***	(0.047)
Social Sciences	0.395***	(0.037)
Law	-0.264***	(0.040)
Education	0.341***	(0.036)
Humanities	-0.117	(0.099)
SCP	0.880***	(0.090)
SCP-Gratuidad	1.014***	(0.067)
Health	-0.063*	(0.033)
Technology	-0.326***	(0.030)
mineduc=3	0.166***	(0.031)
mineduc=10	0.499***	(0.034)
mineduc=20	-0.715***	(0.044)
mineduc=31	0.508***	(0.040)
mineduc=39	0.505***	(0.037)
mineduc=45	0.237***	(0.033)
mineduc=54	0.702***	(0.053)
mineduc=71	0.132***	(0.031)
mineduc=72	0.261***	(0.032)
mineduc=75	0.214***	(0.037)
mineduc=76	0.179***	(0.033)
mineduc=78	0.116***	(0.032)
mineduc=85	0.451***	(0.039)
mineduc=86	-0.002	(0.023)
mineduc=87	0.159***	(0.026)
mineduc=88	-0.293***	(0.027)
mineduc=89	-0.005	(0.027)
mineduc=90	0.029	(0.030)
mineduc=93	0.121***	(0.039)
mineduc=94	0.064	(0.039)
mineduc=990	0.094***	(0.034)
mineduc=991	0.457***	(0.032)
mineduc=992	-0.326***	(0.032)
mineduc=993	0.146***	(0.014)
cut1	-4.826***	(0.084)
cut2	-3.994***	(0.084)
cut3	-2.613***	(0.084)

cut4	-2.111***	(0.084)
Observations	553,563	
Mean of Dep. Variable	3.113	

Standard errors in parentheses

Ordered logistic regression of discretized credit achievement (0-4).Ordered logistic regression of discretized credit achievement (0-4). Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardized. Base category is low SES, mean ability, enrolled in Business at Universidad de Chile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Estimation

C.1 Classification of unobserved types

Following the two-step grouped fixed-effects approach of Bonhomme et al. (2022), as implemented empirically by Joensen and Mattana (2024), we classify enrolled students into M latent types via k-means on two standardized, individual-average residual moments: completion conditional on registered credits and completion relative to the intended annual course load. Both moments are obtained as OLS residuals from

$$y_{it} = \mathbf{x}_i'\alpha_x + \mathbf{z}_{it}'\alpha_z + \lambda_{field(j)} + \lambda_{inst(j)} + \lambda_t + \varepsilon_{it}, \tag{15}$$

where y_{it} is, in turn, the registered-credit completion rate and the intended-annual-load completion rate, $\mathbf{x}_i$ collects individual controls (sex, standardized PSU and GPA, socioeconomic status), and $\mathbf{z}_{it}$ collects state controls (previous-period and cumulative credit completion, delay, and remaining-credits-to-graduate indicators). Fixed effects for field, institution (HEI), and period are included, allowed to differ between SCP and university programs. Residuals are collapsed to individual-level averages and standardized to unit variance before clustering.

Clustering uses k-means++ initialization: cluster centers are seeded sequentially with probability proportional to squared distance from existing centers, and the procedure is repeated over 20 independent seeds to guard against convergence to a local optimum, keeping the run with the lowest within-cluster sum of squares (WCSS). Resulting clusters are relabeled from highest to lowest average of the two standardized moments, so that Type 1 always denotes the highest-productivity group. $M = 2$ is our production classification, entering the structural estimation as an element of Z_i; $M = 3$ is estimated on the same moments purely as a robustness check on this choice of M and is not used in any structural estimate.

Table B6 compares the fit and stability of the $M = 2$ and $M = 3$ classifications, and Table B7 compares the resulting type profiles.

Table B6: Comparison of two- and three-type classifications

K	Students	Moments	WCSS	Q(K)	Explained (%)	CH	Best seed	Optimum starts	Min. share (%)	Max. share (%)
2	191,058	2	130,990	0.686	65.7	366,277	2	20	24.0	76.0
3	191,058	2	81,543	0.427	78.7	352,121	1	20	9.6	67.0

Notes: Both classifications use the same two standardized individual-average residual moments and 20 k-means++ starts. WCSS mechanically decreases with K. K=2 is the production classification; K=3 is appendix-diagnostic only.

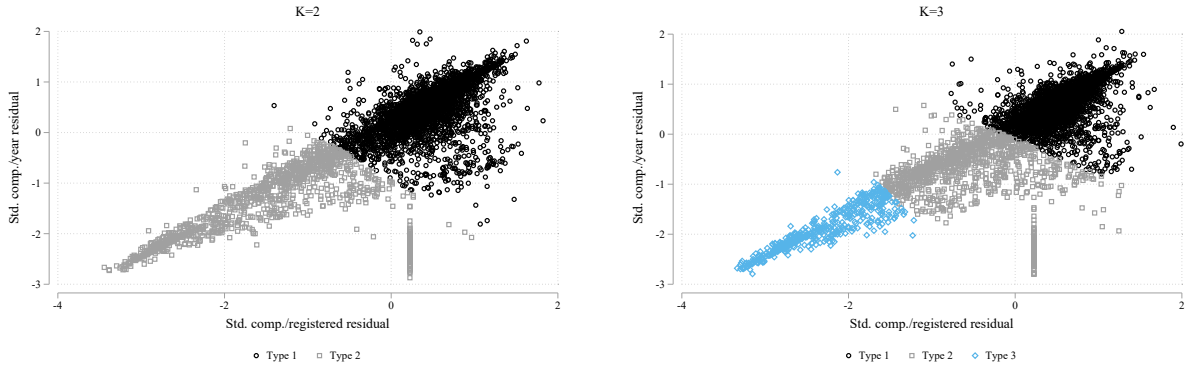
Moving from $M = 2$ to $M = 3$ mechanically lowers WCSS and raises the share of variance

Table B7: Profiles of two- and three-type classifications

	K=2		K=3		
Statistic	Type 1	Type 2	Type 1	Type 2	Type 3
Students	145,202	45,856	128,019	44,621	18,418
Share (%)	76.0	24.0	67.0	23.4	9.6
Std. comp./registered residual	0.448	-1.417	0.525	-0.508	-2.421
Std. comp./year residual	0.463	-1.467	0.576	-0.836	-1.979
Completed/registered ratio	0.902	0.299	0.930	0.560	0.106
Completed/intended-year ratio	0.830	0.211	0.870	0.389	0.078
Graduated by current credit state (%)	38.5	0.0	43.2	1.6	0.0
Behavioral dropout before t=6 (%)	53.6	94.7	49.8	87.9	99.1
Forced terminal exit at t=6 (%)	7.8	5.3	7.0	10.5	0.9
Observed academic periods	3.69	1.47	3.81	2.12	1.11

Notes: Types are ordered from highest to lowest mean standardized academic-productivity residual. Behavioral dropout excludes forced sixth-year nongraduate exit. K=2 is used in all structural estimates; K=3 is diagnostic only.

Figure A11: Classification moments by type: K=2 vs. K=3



a) K=2 b) K=3 (appendix diagnostic)

Notes: Standardized comp./registered residual (x-axis) against standardized comp./year residual (y-axis), by assigned type. A random subsample of 5,000 students is plotted for legibility; the k-means fit itself uses the full classification sample.

explained (Table B6), since additional clusters can only improve within-sample fit. The improvement is not merely mechanical curve-fitting, however: the Calinski-Harabasz index also favors $M = 2$ over $M = 3$ once the loss of degrees of freedom is taken into account, and the additional cluster comes from splitting the low-productivity $M = 2$ group (Type 2, 25.1% of students) into a moderate-productivity group and a very small, very-low-productivity group (Type 3 under $M = 3$, 10.5% of students, completed/registered ratio of 0.107, and essentially universal behavioral dropout before the sixth period). This third group is small enough, and separated enough from the rest of the distribution, that Figure A11 shows the additional structure is visually apparent but concentrated in a thin tail rather than reflecting a qualitatively distinct third mode. We therefore retain $M = 2$ for parsimony and cluster-size balance in the production structural estimates, treating $M = 3$ as an appendix robustness check on the sensitivity of the type classification to M.

C.2 Functional forms

C.2.1 Performance outcomes

For $g \in \mathcal{G} = \{0, 1, 2, 3, 4\}$, denote $\tilde{g}$ the underlying index:

$$\begin{aligned}\tilde{g}_j(x_{it}) = &\sum_m \sum_n \alpha^g_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha^g_{gm} Z_{gm} Z_{in} + \mathbf{Z_g}' \alpha^g_G \\ &+ \alpha^g_{field(j)} + \alpha^g_{inst(j)} + \lambda^g_t + \lambda^g_{tscp} \cdot \text{SCP}_{ij},\end{aligned} \tag{16}$$

where Z_i = (psu, gpa, income, female), Z_j = (field1, …, field10), Z_g = (g_{it-1}, G_{it-1}, delay_{it-1}, 1left_{it-1}, 2left_{it-1}, 3left_{it-1}). g_{it-1} and G_{it-1} denote previous-period and cumulative credit completion, respectively. Xleft indicates whether the student left the program with X or more credits remaining. $\alpha^g_{field(j)}$ are field fixed effects, $\alpha^g_{inst(j)}$ are institution fixed effects, and λ^g_t are period fixed effects. Period fixed effects are allowed to differ between SCPs and university programs.

Xleft variables take into account the fact that passing all the credits in a period where less credits are needed to graduate might be different. An alternative modelling approach would estimate separate models for each possible credit left in a year (1,2,3,4). However, estimating it together gives us more precise estimates.

C.2.2 Dropout probability

We estimate the following logistic regression by maximum likelihood:

$$\begin{aligned} \tilde{drop}_j(x_{it}) = &\sum_m \sum_n \alpha^d_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha^d_{gm} Z_{gm} Z_{in} + \mathbf{Z_g}' \alpha^d_G \\ &+ \alpha^d_{field(j)} + \alpha^d_{inst(j)} + \lambda^d_t + \lambda^d_{tscp} \cdot \text{SCP}_{ij}, \end{aligned} \tag{17}$$

where Z_i = (psu, gpa, income, female), Z_j = (field1, ..., field10), Z_g = (g_{it-1}, G_{it-1}, delay_{it-1}, 1left_{it-1}, 2left_{it-1}, 3left_{it-1}). g_{it-1} and G_{it-1} denote previous-period and cumulative credit completion, respectively. Xleft indicates whether the student left the program with X or more credits remaining. $\alpha^g_{field(j)}$ are field fixed effects, $\alpha^g_{inst(j)}$ are institution fixed effects, and λ^g_t are period fixed effects. Period fixed effects are allowed to differ between SCPs and university programs.

C.2.3 Labour market

We estimate the following wage equation using CASEN survey data,

$$\begin{aligned} \log(w_j(x_{it})) = &\beta^w_0 + \beta^w_{f,field(j)} \,\text{female}_i + \beta^w_e \,\text{exper}_{it} + \beta^w_{e2} \,\text{exper}^2_{it} \\ &+ \sum_{r=1}^{16} \beta^w_{r,field(j)} \, R_{ir} + \beta^w_{inst_type(j),\, field(j)} + \varepsilon_{ijt}, \end{aligned} \tag{18}$$

where R_{ir} is a dummy for region r and $\text{inst_type}(j) \in \{\text{SCP}, \text{Private}, \text{CRUCH}\}$ denotes the institution type of program j. The gender gap and the institution-type premium are both allowed to vary by field. The region-field interactions $\{R_{ir} \, \beta^w_{r,field(j)}\}$ are crucial for identification of the wage coefficient α_w in the structural model, as they serve as an exclusion restriction: regional-field variation affects utility only through wages.

CASEN observes broad institution type but not the specific institution attended. To capture within-type variation in earnings, we separately regress log wages on institution (hei_code) and field fixed effects using institution-level graduate earnings data scraped from the Chilean Ministry of Education's Mifuturo platform. Because part of an institution's Mifuturo fixed effect already reflects its institution type, we demean each institution's fixed effect by subtracting the mean fixed effect across institutions of the same type, so that only the institution's deviation from its type average is retained. We then predict Equation 18 for each student in the sample and add this demeaned institution premium, which avoids double-counting the type-level premium already identified in Equation 18.

C.2.4 Fixed costs

$$FC_j(x_{it}) = \alpha_p OP_{ij} + \sum_m \alpha_{pm}\, OP_{ij} Z_{im} + \sum_m \sum_n \alpha_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha_{gm} Z_{gm} Z_{in} + \mathbf{Z_g}'\alpha_G + \sum_m \sum_k \alpha_{km}\, M_{ij,k} Z_{im} + \mathbf{M_{ij}}'\alpha_m + \alpha_{field(j)} + \alpha_{inst(j)}, \tag{19}$$

where D_{ij} is a dummy for the institution being in a different region (distance proxy), and A_{ij} is the student's relative ability in program j. $M_{ij} = (D_{ij}, A_{ij})$ is a vector of student–program matched variables.

C.3 Conditional Value Function

Conditional value and the terminal boundary. Let $\mathcal{A}_{i,t+1}(\bar{g})$ indicate that, after performance outcome $\bar{g}$, the student has a genuine choice between staying and dropping out. The post-performance continuation value $\bar{V}(x_{i,t+1}(\bar{g}))$ is

$$\bar{V}(x_{i,t+1}(\bar{g})) = \begin{cases} \gamma + \alpha_w wage_0(x_{i,t+1}) - \ln p_{0,t+1}(x_{i,t+1}(\bar{g})), & \mathcal{A}_{i,t+1}(\bar{g}) = 1, \\ \alpha_w wage_j(x_{i,t+1}), & \text{graduation}, \\ \alpha_w wage_0(x_{i,t+1}), & \text{terminal nongraduation}. \end{cases} \tag{20}$$

The final academic period still contains effort and a performance draw. What is absent is a choice after that draw: a nongraduate exits mechanically. Thus the terminal nongraduation branch contains the dropout wage, but neither $-\ln p_0$ nor γ.

For an enrolled alternative,

$$v_j(x_{it}, e_{it}) = -FC_j(x_{it}) - c_j(x_{it})e_{it} + \beta \sum_{\bar{g}\in\mathcal{G}} \phi_{ijt}^{\bar{g}}(e_{it})\bar{V}(x_{i,t+1}(\bar{g})). \tag{21}$$

At an interior effort optimum, the first-order condition implies

$$v_j(x_{it}, e_{ijt}^*) = -FC_j(x_{it}) + \beta \sum_{\bar{g}\in\mathcal{G}} a_{ijt}^{\bar{g}} \bar{V}(x_{i,t+1}(\bar{g})), \tag{22}$$

$$a_{ijt}^{\bar{g}} \equiv \phi_{ijt}^{\bar{g}}(e_{ijt}^*) - e_{ijt}^* \left.\frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\right|_{e_{it}=e_{ijt}^*}. \tag{23}$$

Separating the part whose coefficient is fixed at β from the wage component whose coefficient identifies α_w gives

$$EMAX_{ijt} = \sum_{\bar{g}\in\mathcal{G}} a_{ijt}^{\bar{g}} \mathbb{1}\{\mathcal{A}_{i,t+1}(\bar{g}) = 1\} \left[\gamma - \ln p_{0,t+1}(x_{i,t+1}(\bar{g}))\right], \tag{24}$$

$$EWAGE_{ijt} = \beta \sum_{\bar{g}\in\mathcal{G}} a_{ijt}^{\bar{g}} wage_{\bar{g}}(x_{i,t+1}), \tag{25}$$

$$v_j(x_{it}, e_{ijt}^*) = -FC_j(x_{it}) + \beta EMAX_{ijt} + \alpha_w EWAGE_{ijt}. \tag{26}$$

Here $wage_{\bar{g}} = wage_j$ on graduation branches and $wage_{\bar{g}} = wage_0$ otherwise. In the implementation's terminal period $t = 6$, the active-choice indicator is zero for every outcome, so $EMAX_{ijt} = 0$ while $EWAGE_{ijt}$ remains outcome-dependent. where $\Lambda(\cdot)$ denotes the standard logistic CDF. The general expression for ordered logit probabilities are, denoting

by $\alpha_{\bar{g}}$ the estimated cutoff points,

$$\begin{aligned}\phi_{ijt}^{\bar{g}}(e_j^*(x_{it})) &= \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \\ &= \frac{1}{1+\exp(e_j^*(x_{it}) - \alpha_{\bar{g}})} - \frac{1}{1+\exp(e_j^*(x_{it}) - \alpha_{\bar{g}-1})}\end{aligned}$$

The derivative wrt $e_j^*(x_{it})$ writes

$$\begin{aligned}\frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} &= -\Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it}))) \\ &\quad - \big(-\Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})))\big) \\ \frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} &= \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it}))) \\ &\quad - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})))\end{aligned}$$

Computing the marginal-cost and envelope expression gives

$$\sum_{\bar{g}\in\mathcal{G}} a_{ijt}^{\bar{g}} \bar{V}(x_{i,t+1}(\bar{g})).$$

The ordered-logit expansions below describe the weights $a_{ijt}^{\bar{g}}$. Whenever they display the bracket $\alpha_w wage_0 - \ln p_0$, that bracket applies only to an active nongraduation branch. Graduation and terminal nongraduation instead use the wage-only branches in Equation 20; their contribution to $EMAX$ is zero and their wage contribution remains in $EWAGE$.

For $g = 1$

$$\begin{aligned}&\big(\Lambda(\alpha_1 - e_j^*(x_{it})) + \big(\Lambda(\alpha_1 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_1 - e_j^*(x_{it})))\big) \times e_j^*(x_{it})\big) \\ &\big(\Lambda(\alpha_1 - e_j^*(x_{it})) + \big(\Lambda(\alpha_1 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_1 - e_j^*(x_{it})))\big) \times e_j^*(x_{it})\big)\end{aligned}$$

$$\begin{aligned}&\big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(1))\big) \times \\ &\left(\Lambda(\alpha_1 - index_j(x_{it})) - \frac{\partial \phi_{ijt}^{\bar{g}}(e_j(x_{it}))}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} \times e_j^*(x_{it})\right)\end{aligned}$$

We defined effort as

$$e_j(x_{it}) = \frac{1 - \Lambda(\alpha_1 - index_j(x_{it}))}{\Lambda(\alpha_1 - index_j(x_{it}))} \tag{27}$$

$$\Lambda(\alpha_1 - index_j(x_{it})) = \frac{1}{1 + e_j(x_{it})} \tag{28}$$

$$\alpha_1 - index_j(x_{it}) = \ln\left(\frac{\frac{1}{1+e_j(x_{it})}}{1 - \frac{1}{1+e_j(x_{it})}}\right) \tag{29}$$

$$\alpha_1 - index_j(x_{it}) = \ln\left(\frac{1}{e_j(x_{it})}\right) = -\ln(e_j(x_{it})) \tag{30}$$

$$\alpha_1 - \ln(e_j(x_{it})) = index_j(x_{it}) \tag{31}$$

then the derivative wrt effort looks like

$$\left(\Lambda(\alpha_1 - index_j(x_{it})) - \left(\lambda(\alpha_1 - index_j(x_{it})) \times \frac{-1}{e_j(x_{it})}\right) \times e_j(x_{it})\right)$$
$$= \frac{1}{1 + e_j(x_{it})} + \frac{e_j(x_{it})}{(1 + e_j(x_{it}))^2}$$

For $g \in (2, 4)$

$$\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times$$
$$(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it}))-$$
$$(\Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it}))) - \Lambda(\alpha_{\bar{g}} - index_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - index_j(x_{it}))))$$

Recall that rearranging the effort measure in Equation 27, we can express $index_j(x_{it}) = \alpha_1 - \ln(e_j(x_{it}))$

$$
\begin{aligned}
&\big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\big) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&\frac{\partial \phi^{\bar{g}}_{ijt}(e_j(x_{it}))}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})} (\Lambda(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it}))) - \Lambda(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))) \times e^*_j(x_{it})\Bigg) \\
&= \big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\big) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&\left(\lambda(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it})) \times \left(\frac{-1}{e^*_j(x_{it})}\right) - \lambda(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))) \times \left(\frac{-1}{e^*_j(x_{it})}\right)\right) \times e^*_j(x_{it})\right) \\
&= \big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\big) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&\left(\frac{-e^*_j(x_{it})}{e^*_j(x_{it})}\right) \times \left(\frac{\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it}))))^2} - \frac{\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))))^2}\right)\Bigg) \\
&= \big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\big) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) + \\
&\left(\frac{\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it}))))^2} - \frac{\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))))^2}\right)\Bigg)
\end{aligned}
$$

For $g = 5$

$$
\begin{aligned}
&\big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(5))\big) \times \\
&\big(1 - \Lambda(\alpha_4 - e^*_j(x_{it})) - \big(\Lambda(\alpha_4 - e^*_j(x_{it})) \times (1 - \Lambda(\alpha_4 - e^*_j(x_{it})))\big) \times e^*_j(x_{it})\big)
\end{aligned}
$$

$$
\sum_{\bar{g}\in\mathcal{G}} \left(\phi^{\bar{g}}_{ijt}(e^*_j(x_{it})) - \frac{\partial \phi^{\bar{g}}_{ijt}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})} e^*_j(x_{it})\right)
$$

Notice from the above expression that

$$
\sum_{\bar{g}\in\mathcal{G}} \big(\phi^{\bar{g}}_{ijt}(e^*_j(x_{it}))\big) = 1
$$

Since we are integrating over all possible states and that

$$
\begin{aligned}
&\sum_{\bar{g}\in\mathcal{G}}\left(\frac{\partial\phi^{\bar{g}}_{ijt}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})}e^*_j(x_{it})\right)\\
&=\lambda(\alpha_1-index(x_{it})\times(-1/e^*_j(x_{it}))\times e^*_j(x_{it})+\\
&(\lambda(\alpha_2-index(x_{it}))-\lambda(\alpha_1-index(x_{it})))\times(-1/e^*_j(x_{it}))\times e^*_j(x_{it})+\\
&(\lambda(\alpha_3-index(x_{it})-\lambda(\alpha_2-index(x_{it})))\times(-1/e^*_j(x_{it}))\times e^*_j(x_{it})+\\
&(\lambda(\alpha_4-index(x_{it})-\lambda(\alpha_3-index(x_{it})))\times(-1/e^*_j(x_{it}))\times e^*_j(x_{it})+\\
&(-\lambda(\alpha_4-index(x_{it})))\times(-1/e^*_j(x_{it}))\times e^*_j(x_{it})
\end{aligned}
$$

since $index=\ln(e^*_j(x_{it}))+\alpha_1$ and that $\frac{\partial\lambda(\alpha_g-index(x_{it}))}{\partial e_{it}}=\lambda(\alpha_g-index(x_{it}))\times(-1/e^*_j(x_{it}))$.

This can be rewritten as

$$
\begin{aligned}
&\sum_{\bar{g}\in\mathcal{G}}\left(\frac{\partial\phi^{\bar{g}}_{ijt}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})}e^*_j(x_{it})\right)\\
&=-\lambda(\alpha_1-index(x_{it})+\\
&-(\lambda(\alpha_2-index(x_{it}))-\lambda(\alpha_1-index(x_{it})))+\\
&-(\lambda(\alpha_3-index(x_{it})-\lambda(\alpha_2-index(x_{it})))+\\
&-(\lambda(\alpha_4-index(x_{it})-\lambda(\alpha_3-index(x_{it})))+\\
&-(-\lambda(\alpha_1-index(x_{it})))\\
&=0
\end{aligned}
$$

Period and alternative conventions. The outside option is terminal in every period:

$$v_0(x_{it}) = \alpha_w wage_0(x_{it}),$$

with no discount factor multiplying this contemporaneous terminal payoff. For an enrolled alternative before the terminal boundary, the active branches use the Hotz–Miller expression in Equation 20. In the final academic period, effort still changes the probability of graduation, but the continuation value after performance is simply

$$\bar{V}(x_{i,T+1}(\bar{g})) = \begin{cases} \alpha_w wage_j(x_{i,T+1}), & \text{graduation,} \\ \alpha_w wage_0(x_{i,T+1}), & \text{nongraduation.} \end{cases}$$

Thus terminal $EMAX$ is zero, whereas terminal $EWAGE$ is generally nonzero.

C.4 Estimation algorithm

We estimate the model jointly using an expectation-maximization (EM) algorithm in the spirit of Arcidiacono and Miller (2011), in which the type posteriors are updated using both performance and choices. Two features matter for the sample. First, the trait distribution $\pi_m(Z_i^{obs})$ is a *population* prior conditional on observed characteristics, and it is defined for every student in the structural sample, including those who never enroll. Individuals who choose the outside option have no performance or dropout history, so they inform the type distribution only through their enrollment decision, which is what allows the type shares to be identified in the population rather than among enrollees. Second, students observed in the panel of enrolled students but outside the structural sample have no enrollment choice set; for them the posterior is updated from performance and dropout alone, with a prior conditional on enrollment. The algorithm proceeds as follows.

1. **Wages.** We estimate the wage equation Equation 11 on survey data and predict graduation and dropout wages for every program-state. Since the survey is not linked to students' higher education histories, wages do not depend on the unobserved trait and can be estimated once, outside the EM loop.

2. **Starting values.** The EM algorithm requires starting values for the posteriors q_{im}. We start from posteriors that increase with students' average performance, and verify that the converged estimates do not depend on this choice by repeating the estimation from alternative starts: a preliminary two-step classification of types, the same start with the type labels reversed, and random posteriors.

3. **EM loop.** Iterate over the following steps until the posteriors, the parameters and the value of the objective stop changing:

   (a) *Reduced-form components (performance and dropout CCPs), M-step.* Weighting each student by the current posterior q_{im}, we estimate the ordered logit of credit completion (0–4), which delivers the index $\ln e_j^*(x_{it})$, and the dropout logit $p_{0,t+1}$. Both include program fixed effects, period fixed effects that differ between SCPs and universities, and trait shifters. They are estimated on the *panel sample*, which contains all enrolled students and is considerably larger than the structural sample, so they are estimated more precisely. This step delivers, for each student and type, the likelihood L_{im}^{perf} of the observed sequence of performance outcomes. The dropout logit is used differently: it is not a likelihood term for students in the structural sample, whose observed continuation decisions are explained by the structural model below. Its role there is to supply the predicted dropout probabilities $p_{0,t+1}$ at the states of the structural data, for each type.

   (b) *Continuation values.* Evaluating the estimated dropout probabilities at every state of the structural data, we obtain $\bar{V}(x_{i,t+1}(\bar{g}))$ through Equation 12 separately for each type, and hence $EMAX_{ijt}^m$, $EWAGE_{ijt}^m$ and the marginal costs $c_j^m(x_{it})$ through Equation 13.

   (c) *Fixed costs, utility parameters and type prior, M-step.* Taking the continuation values as given, we maximize over the fixed-cost and utility parameters θ (including FC_j and α_w) and over the parameters λ of the type prior $\pi_m(Z_i^{obs};\lambda)$, a multinomial logit in observed characteristics, the mixture likelihood of all students in the *structural choice sample*, enrolled or not:

$$\sum_i \ln \sum_m \pi_m(Z_i^{obs};\lambda)\, L_{im}^{perf} \prod_t P_{imt}(\theta),$$

   where $P_{imt}(\theta)$ is the probability of student i's observed choice at t given type m: the program or outside-option choice at $t = 0$, based on v_j in Equation 14, and the continue-or-exit decision afterwards. The dropout CCP is not included in L_{im}^{perf}, because the continue-or-exit decision already enters through P_{imt} and would otherwise be counted twice. Program fixed effects are included in FC_j, so that fixed costs rationalize the observed enrollment shares across programs, while the sensitivity to out-of-pocket fees is identified from the variation in public funding eligibility.

(d) *Type updating, E-step.* For students in the structural sample, the posterior is

$$q_{im} = \frac{\pi_m(Z_i^{obs}; \lambda)\, L_{im}^{perf} \prod_t P_{imt}(\theta)}{\sum_{m'} \pi_{m'}(Z_i^{obs}; \lambda)\, L_{im'}^{perf} \prod_t P_{im't}(\theta)},$$

so that it uses the program choice as well as performance, and never-enrollees are assigned a posterior that reflects their characteristics and their decision not to enroll. For students in the panel who are not in the structural sample, we use

$$q_{im} \propto \pi_m^E(Z_i^{obs}, tier_i)\, L_{im}^{perf}\, L_{im}^{drop},$$

where L_{im}^{drop} is the likelihood of the observed continuation or dropout decisions and π_m^E is a multinomial logit of the type on observed characteristics and the tier of the first program, fitted with the current posteriors. Since these students are enrollees, π_m^E describes the type mix conditional on enrollment and program, not the population prior. Students observed for more periods have more informative posteriors, and the posteriors are carried over as weights to the next iteration.

4. **Marginal costs.** Once the algorithm has converged, we use the final CCPs, performance estimates and posteriors to compute $c_j(x_{it})$ from the FOC Equation 13, stacking each realized higher education episode once per type and weighting it by q_{im}.